REGULARITY OF PLURISUBHARMONIC EXHAUSTIONS AND RELATED PROBLEMS

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Abstract. We construct quasi-projective and non-Kähler threefolds with locally Lipschitz plurisubharmonic (psh) exhaustions but no C^1 psh exhaustion, answering Nakano's smoothing question negatively in dimensions at least three. We prove an affirmative result for divisor complements in compact Kähler surfaces. For line bundles over compact complex manifolds, we characterize the existence of psh exhaustions of the dual total space by semipositive metrics of the same C^r, Hölder, or real analytic regularity, establishing the seminegative converse discussed by Ohsawa. We also prove a holomorphic convexity criterion for reduced complex spaces fibered by curves over affine space, extending Vîjîitu's curve-fibering argument to singular spaces and giving a higher-dimensional analogue of Ohsawa's surface theorem.

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1. Introduction to the main results and applications

A *continuous plurisubharmonic (psh) exhaustion* of a reduced complex space X is a continuous function $\rho : X \to \mathbb{R}$ whose pullback along every holomorphic disc is subharmonic and whose sublevel sets $\{\rho \leq c\}$ are compact for all $c \in \mathbb{R}$. We call X *weakly* 1*-complete*, or *weakly pseudoconvex*, if it admits a smooth psh exhaustion.

Can smoothness be replaced by continuity in this definition? Nakano asked this in 1980, calling manifolds with a continuous psh exhaustion *Lelong-pseudoconvex* [Wak80, §I.1, Problem 1, p. 162]. Demailly's thesis allows any psh exhaustion in the definition of weak pseudoconvexity [Dem82b, Introduction, §1, p. 3]. On a Kähler manifold, a continuous psh exhaustion already gives a complete Kähler metric, as needed for his L^2 estimates for solving $\bar{\partial}u = f$ [Dem82a, Theorems 1.3 and 4.1].

1.1. Main Results.

In the following theorem we construct a quasi-projective counterexample.

Theorem 1.1. *There exists a connected smooth quasi-projective Kähler threefold M admitting a psh exhaustion of class $C^{0,1}_{\mathrm{loc}}$ but no psh exhaustion of class $C^{1,0}$.*

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Here by $C^{k,\alpha}_{\mathrm{loc}}$, for $k \geq 0$ an integer and $0 < \alpha \leq 1$, we mean C^k functions whose derivatives of order k are locally α-Hölder in real coordinates. Thus $C^{0,1}_{\mathrm{loc}}$ means locally Lipschitz. At exponent zero we write $C^{k,0} = C^k$, so $C^{1,0}$ means continuously differentiable.

The construction uses Grauert's ruled-surface example [Gra62, §4, 8(d)] and Koike's variant [Koi21, Example 4.1]. In each case the ruled surface S has a negative section Y and a distinguished line bundle L. This bundle is nef and big (Proposition 2.1) and admits a locally Lipschitz semipositive metric (Proposition 2.6). By Proposition 2.3, no neighborhood of Y carries a C^1 semipositive metric on L. A Hermitian metric h is semipositive if its weights $\varphi = -\log \|e\|_h^2$ in local holomorphic frames e are psh. We take $M = \mathrm{Tot}(L^{-1})$. By the regularity criterion (Theorem 3.4),

$$M \text{ has a psh exhaustion of class } \mathcal{R}$$
$$\iff \quad L \text{ has a semipositive metric of class } \mathcal{R}.$$

This equivalence holds for every holomorphic line bundle over a compact complex manifold and every C^r, local Hölder, or real analytic (C^ω) class $\mathcal{R}$ in (3.1). The metric obstruction on L therefore rules out every C^1 psh exhaustion of M.

Taking the product of M with a Hopf surface H gives a five-dimensional non-Kähler counterexample. Since H is compact, projection to M pulls its locally Lipschitz psh exhaustion back to an exhaustion of $M \times H$. A C^1 psh exhaustion would restrict to one on the closed slice $M \times \{h\}$, and a Kähler form would restrict to one on $\{m\} \times H$. To obtain a non-Kähler counterexample in dimension three, we use a modification with a compact center.

Theorem 1.2. *Let M be the threefold in Theorem 1.1. For every $x \in M$ there is a smooth non-Kähler threefold N and a proper holomorphic modification $\mu : N \to M$ such that*

$$N \setminus \mu^{-1}(x) \simeq M \setminus \{x\}.$$

The threefold N admits a psh exhaustion of class $C^{0,1}_{\mathrm{loc}}$ but no psh exhaustion of class $C^{1,0}$.

We first blow up x and choose a line and a conic in the exceptional plane. Following Hironaka's construction [Bar15, §6], we blow up these curves in opposite orders near their two intersection points. Two exceptional rational curves then have homology classes whose sum is zero, which rules out a Kähler form on N. The locally Lipschitz psh exhaustion of M pulls back to N. Conversely, a C^1 psh exhaustion of N could be made constant near $\mu^{-1}(x)$ by composing it with a smooth convex nondecreasing function tending to infinity. It would then descend to a C^1 psh exhaustion of M (Proposition 4.1).

It remains open whether every Lelong-pseudoconvex complex manifold of dimension two is weakly 1-complete. The next theorem rules out counterexamples among complements of divisors in compact Kähler surfaces and, in particular, among smooth quasi-projective surfaces.

Theorem 1.3. *Let S be a smooth compact connected Kähler surface and $D \neq 0$ an effective divisor. If $S \setminus \operatorname{supp} D$ admits a continuous psh exhaustion, it admits a smooth one. Consequently, the same implication holds for every smooth complex quasi-projective surface.*

Our proof strategy is to construct a smooth exhaustion of $X = S \setminus \operatorname{supp} D$ in two cases. If X is a modification of a Stein space, we pull back a smooth strictly psh exhaustion from that space. Otherwise we average positive $\partial\bar\partial$-closed currents supported on levels of an exhaustion. The resulting current has zero Lelong numbers and compact support in X. Using the current theorem of Dinh–Nguyên–Sibony [DNS22] and the Hodge index theorem, we show that there exists a divisor $A = \sum_i a_i D_i$ with positive integer coefficients such that

$$\operatorname{supp} A = \operatorname{supp} D, \qquad A^2 = 0, \qquad \mathcal{O}_S(A) \text{ is unitary flat near } D.$$

Let s_A be the canonical section of $\mathcal{O}_S(A)$ and h a flat Hermitian metric on this bundle near $\operatorname{supp} D$. The function $-\log \|s_A\|_h^2$ is pluriharmonic on a punctured neighborhood of $\operatorname{supp} D$ and tends uniformly to $+\infty$ on approach to $\operatorname{supp} D$. This is the *pluriharmonic logarithmic collar* from which we construct the smooth exhaustion. Section 5 also gives separate numerical and neighborhood criteria for existence of an exhaustion.

For a single smooth connected boundary curve $C \subset S$, the self-intersection $C^2 = \deg N_{C/S}$ and the holomorphic functions on $X = S \setminus C$ are related by

$$C^2 < 0 \quad \Longrightarrow \quad \mathcal{O}(X) = \mathbb{C} \quad \Longrightarrow \quad C^2 \leq 0.$$

Indeed, when $C^2 < 0$, Grauert's contraction criterion [Gra62, §4, 8(e)] identifies X with the complement of a point in a compact normal surface, and every holomorphic function extends across that point. When $C^2 > 0$, the curve C is nef and $(K_S - mC) \cdot C < 0$ for m large. Serre duality and Riemann–Roch then give $h^0(S, \mathcal{O}_S(mC)) > 1$. Dividing a section independent of s_C^m by s_C^m gives a nonconstant holomorphic function on X.

Ohsawa proved that a connected noncompact complex surface admitting a smooth psh exhaustion is holomorphically convex precisely when it has a nonconstant holomorphic function [Ohs81, Proposition 1.4, p. 156]. The following analogue for spaces fibered by curves assumes only a continuous psh exhaustion. For a complement $S \setminus C$ admitting such an exhaustion, it gives holomorphic convexity when $C^2 > 0$, and also when $C^2 = 0$ if the complement carries a nonconstant holomorphic function.

Theorem 1.4. *Let X be a reduced complex space of pure dimension $n \geq 2$ with a continuous real-valued psh exhaustion. If a holomorphic map*

$$F : X \longrightarrow \mathbb{C}^{n-1}$$

has every nonempty fiber of dimension at most one, then X is holomorphically convex and admits a smooth psh exhaustion.

Analytic fiber cycles and Bishop compactness establish the stability needed for the convexity argument.

Throughout this paper, *regularization* means finding a smooth psh exhaustion on a space that admits a continuous one. Among smooth quasi-projective manifolds, Theorems 1.1 and 1.3 place the least possible counterexample dimension at three.

1.2. Applications to earlier problems.

(i) Nakano's smoothing question. As recalled above, Nakano asked whether every Lelong-pseudoconvex manifold is weakly 1-complete. Choose a locally Lipschitz psh exhaustion $\rho \geq 0$ on the threefold M of Theorem 1.1. For each $n > 3$, the product $M \times \mathbb{C}^{n-3}$ is quasi-projective and admits the locally Lipschitz psh exhaustion

$$\rho_n(x, z) = \rho(x) + \|z\|^2.$$

A C^1 psh exhaustion on this product would restrict to one on the closed slice $M \times \{0\}$, contradicting Theorem 1.1. Thus the answer is negative in every dimension at least three. Theorem 1.3 gives an affirmative answer for surfaces with a divisorial compactification in a compact Kähler surface, including smooth quasi-projective surfaces.

(ii) Filip–Tosatti and the bounded-potential question. Filip and Tosatti asked to prove that the distinguished nef and big line bundle in Grauert's ruled-surface example is not smoothly semipositive [FT18, Problem 2.2, p. 2989]. Zhang proved this for the original example using Koike's first obstruction class [Zha25, Theorem 1.2]. He also recorded a question suggested by Tosatti: for the original or modified Grauert example, does $c_1(L)$

contain a closed positive current with bounded local potentials and minimal singularities [Zha25, §4]? Proposition 2.6 and Remark 2.7 answer this affirmatively for both examples. The metric has locally Lipschitz weights and is smooth with strictly positive curvature on $S \setminus Y$. Its curvature has bounded local potentials and minimal singularities. In the notation of (2.2), the original and modified examples have

$$(Y^2, L^2) = (-1, 1), \qquad (Y^2, L^2) = (-2, 2),$$

respectively.

(iii) Koike's first obstruction at C^1 regularity. Koike extended Ueda's obstruction theory to line bundles. His Theorem 1.4 excludes smooth semipositivity when the first obstruction class is nonzero, and Example 4.1 applies this to the modified Grauert example [Koi21]. For the Grauert triples (Y, S, L) considered here, the nonzero extension class ξ in (2.1) gives

$$u_1(Y, S, L) = -\xi \neq 0 \quad \text{in } H^1(Y, N_{Y/S}^{-1}) \simeq H^1(B, F).$$

Proposition 2.3 shows that no neighborhood of Y carries a C^1 Hermitian metric on L with psh weights; the same holds for every positive tensor power of L. Indeed, Lemma 2.2 makes the first normal derivatives of hypothetical C^1 weights holomorphic, and their transition identities would make ξ a coboundary.

(iv) Ohsawa–Vîjîitu convexity. For smooth sources, the proof of Theorem 1.4 follows Vîjîitu's work [Vîj22b, Corollary 4.2 and pp. 463–464]. Our proof extends the fiber-stability argument to singular total spaces. For irreducible surfaces, the continuous-exhaustion case already follows from the stronger theorem [Vîj22a, Theorem 1.1].

(v) Ohsawa's seminegative converse. Ohsawa observed that the total space of a seminegative holomorphic line bundle over a compact complex manifold is weakly 1-complete and discussed the converse [Ohs21, §6, Proposition 6.1 and the following paragraph]. Ohsawa's Theorem 6.1 treats affine line bundles over compact Riemann surfaces. For linear line bundles, taking $\mathcal{R} = C^\infty$ in Theorem 3.4 proves the converse over any compact complex manifold. The criterion also preserves the C^r, local Hölder, and real analytic classes in (3.1).

2. Distinguished line bundles on Grauert's ruled surface

The two regularity properties of the distinguished bundle L come from its behavior along the negative section Y. A C^1 semipositive metric would split the extension defining the ruled surface. On the other hand, the negativity of Y allows us to construct a semipositive metric whose weights are locally Lipschitz across Y.

2.1. **Grauert's construction and its line bundle.** Let B be a smooth connected projective curve of genus $g \geq 2$, let F be a line bundle of degree $d > 0$, and choose $0 \neq \xi \in H^1(B, F)$. Let

$$0 \longrightarrow F \longrightarrow E \longrightarrow \mathcal{O}_B \longrightarrow 0 \tag{2.1}$$

be an extension with class ξ. We write $\mathbf{P}(E) = \mathbf{P}_{\text{lines}}(E)$ for the bundle of lines in E, with tautological line bundle $\mathcal{O}_{\mathbf{P}(E)}(-1)$. Set

$$S = \mathbf{P}(E), \qquad \pi : S \to B, \qquad Y = \mathbf{P}(F), \qquad L = \mathcal{O}_S(Y) \otimes \pi^* F. \tag{2.2}$$

We identify the section Y with B via π. By GAGA, E is algebraic and S is a smooth projective surface. The choice $F = K_B$ works for every $g \geq 2$, since Serre duality gives

$$\deg K_B = 2g - 2 > 0, \qquad H^1(B, K_B) \simeq H^0(B, \mathcal{O}_B)^* \simeq \mathbb{C}.$$

In genus two we may also take $F = \mathcal{O}_B(p)$ for $p \in B$. A nonconstant meromorphic function with at most a simple pole at p would give a degree-one map $B \to \mathbf{P}^1$. Thus $h^0(B, \mathcal{O}_B(p)) = 1$, and Riemann–Roch gives $h^1(B, \mathcal{O}_B(p)) = 1$ as well. These two degree choices recover Grauert's example for $F = \mathcal{O}_B(p)$ [Gra62, §4, 8(d)] and Koike's variant for $F = K_B$ [Koi21, Example 4.1].

Choose local frames f_i of F on $U_i \subset B$ and lifts v_i of $1 \in \mathcal{O}_B$, with

$$f_j = g_{ij} f_i, \qquad v_j = v_i + c_{ij} f_i.$$

The cocycle $\{v_j - v_i = c_{ij} f_i\}$ represents ξ. On $S \setminus Y$, write a line as $[t_i f_i + v_i]$. Near Y, write it as $[f_i + w_i v_i]$, with $w_i = 1/t_i$ where both coordinates are defined. Comparing coefficients on overlaps gives

$$t_i = g_{ij} t_j + c_{ij}, \qquad w_i = \frac{w_j}{g_{ij} + c_{ij} w_j}, \qquad w_j = \frac{g_{ij} w_i}{1 - c_{ij} w_i}. \tag{2.3}$$

The quotient map $E \to \mathcal{O}_B$ induces a section of $\mathcal{O}_S(1)$ whose zero divisor is Y. Consequently

$$\begin{aligned} \mathcal{O}_S(Y) \simeq \mathcal{O}_S(1), \qquad & N_{Y/S} \simeq \operatorname{Hom}(F, E/F) \simeq F^{-1}, \\ Y^2 = -d, \qquad & L|_Y \simeq F^{-1} \otimes F \simeq \mathcal{O}_Y. \end{aligned} \tag{2.4}$$

Let s_Y be the canonical section of $\mathcal{O}_S(Y)$. Near Y, let d_i be the frame defined by $s_Y = w_i d_i$, and put $\ell_i = d_i \otimes \pi^* f_i$. Then

$$d_j = \frac{1 - c_{ij} w_i}{g_{ij}} d_i, \qquad \ell_j = (1 - c_{ij} w_i) \ell_i. \tag{2.5}$$

In particular, the frames $\ell_i|_Y$ agree and give the trivialization in (2.4).

Proposition 2.1. *The line bundle L is nef and big, with*

$$L^2 = d, \qquad L \cdot Y = 0, \qquad \operatorname{Null}(L) = Y.$$

The bundle $A = L^{\otimes 2} \otimes \mathcal{O}_S(-Y)$ is ample. Every section of a positive tensor power of L vanishes along Y. In particular, L is not semiample.

Proof. Let P be a fiber of π. The intersection identities are

$$Y^2 = -d, \quad Y \cdot \pi^* F = d, \quad (\pi^* F)^2 = 0, \quad Y \cdot P = 1, \quad \pi^* F \cdot P = 0.$$

Thus $L^2 = d$, $L \cdot Y = 0$, and $L \cdot P = 1$. For an irreducible horizontal curve $D \neq Y$, let $n = \deg(\pi|_{\widetilde{D}}) > 0$, where $\widetilde{D}$ is its normalization. Then

$$L \cdot D = Y \cdot D + nd > 0.$$

Thus L is nef, and $L^2 = d > 0$ makes it big by the numerical criterion for nef bundles on a projective surface. The only curve on which L has degree zero is Y. For A we have

$$\begin{gathered} A = \mathcal{O}_S(Y) \otimes \pi^*(F^{\otimes 2}), \qquad A^2 = 3d, \\ A \cdot Y = d, \qquad A \cdot P = 1, \qquad A \cdot D = Y \cdot D + 2nd > 0. \end{gathered}$$

The Nakai–Moishezon criterion now gives ampleness.

To prove the assertion about sections, we show that $S \setminus Y$ contains no compact irreducible curve. Such a curve D would have to dominate B, since each fiber in the complement is an affine line. Its normalization would give a finite map $p : \widetilde{D} \to B$ of degree $n > 0$. Away from the branch values, put

$$a_i(b) = \frac{1}{n} \sum_{x \in p^{-1}(b)} t_i(x).$$

The sum is holomorphic and single-valued, and is locally bounded at branch values by compactness of the inverse image of a small closed disc. It extends holomorphically across them. Equation (2.3) implies

$$a_i = g_{ij}a_j + c_{ij}, \qquad c_{ij}f_i = a_i f_i - a_j f_j.$$

This makes the cocycle representing ξ a coboundary, a contradiction.

Now let $\sigma \in H^0(S, L^{\otimes m})$, $m > 0$. Its restriction to Y is constant. If that constant were nonzero, the zero divisor of σ would be disjoint from Y and would be nonempty, since

$$L^{\otimes m}|_P \simeq \mathcal{O}_{\mathbf{P}^1}(m), \qquad \deg \operatorname{div}(\sigma|_P) = m > 0.$$

It would be a compact curve in $S \setminus Y$, which is impossible. Hence $\sigma|_Y = 0$ for every $m > 0$. □

2.2. **The first normal derivative and the C^1 obstruction.** On Y, the weights of a semipositive metric on L glue to a subharmonic function and hence are constant in the frames ℓ_i. The next lemma determines what plurisubharmonicity then imposes on their first derivatives normal to Y.

Lemma 2.2 (First normal derivative)**.** *Let $D \subset \mathbb{C}$ be a disc and let φ be a real-valued C^1 psh function on a neighborhood of $D \times \{0\}$ in $\mathbb{C}^2$. If $\varphi(z, 0)$ is constant, then $b(z) = \partial_w \varphi(z, 0)$ is holomorphic.*

Proof. Subtract the constant $\varphi(z, 0)$. By continuity of $\partial_w \varphi$, the first-order expansion

$$\varphi(z, w) = 2\operatorname{Re}(b(z)w) + o(|w|) \tag{2.6}$$

is uniform on each compact subdisc in the z variable. Fix a holomorphic function f on D. On each subdisc $D_0 \Subset D$ and for all sufficiently small $t > 0$, the graph restrictions

$$u_t^{\pm}(z) = \frac{\varphi(z, \pm t f(z))}{t}$$

are subharmonic and converge uniformly to $\pm 2\operatorname{Re}(bf)$ by (2.6). Both limits are subharmonic, so $\operatorname{Re}(bf)$ is harmonic on D. Taking $f = 1, i, z, iz$ shows that b and zb are harmonic. It follows that

$$0 = \Delta(zb) = z\Delta b + 4\partial_{\bar z} b.$$

Thus $\partial_{\bar z} b = 0$. □

Proposition 2.3. *No neighborhood of Y carries a C^1 Hermitian metric on L with psh weights. The same holds for $L^{\otimes m}$, $m \geq 1$.*

Proof. For a Hermitian metric h on a line bundle A, its weights $\varphi_i = -\log \|e_i\|_h^2$ in local frames $e_j = g_{ij}e_i$ satisfy

$$\varphi_j = \varphi_i - \log |g_{ij}|^2. \tag{2.7}$$

With $d^c = \frac{i}{4\pi}(\bar\partial - \partial)$ and $dd^c = \frac{i}{2\pi}\partial\bar\partial$, the curvature is $\Theta_h(A) = dd^c\varphi_i$, in the sense of currents for weights of regularity below C^2. Suppose the weights φ_i in the frames ℓ_i are C^1 and psh near Y. Applying (2.7) to (2.5) gives

$$\varphi_j(z, w_j) = \varphi_i(z, w_i) - \log |1 - c_{ij}w_i|^2. \tag{2.8}$$

The restrictions $\varphi_i|_Y$ glue to a subharmonic function on the compact connected curve Y, hence to a constant. Rescale the metric so that $\varphi_i(z, 0) = 0$. By Lemma 2.2, the functions

$$b_i(z) = \partial_{w_i}\varphi_i(z, 0)$$

are holomorphic. Differentiating (2.8) at $w_i = 0$ and using (2.3) gives

$$g_{ij}b_j = b_i + c_{ij}, \qquad v_j - b_j f_j = v_i + (c_{ij} - g_{ij}b_j)f_i = v_i - b_i f_i.$$

Therefore

$$s_i : \mathcal{O}_B|_{U_i} \longrightarrow E|_{U_i}, \qquad a \longmapsto a(v_i - b_i f_i)$$

glue to a holomorphic right inverse of $E \to \mathcal{O}_B$, contradicting $\xi \neq 0$. A C^1 semipositive metric on $L^{\otimes m}$ would give one on L by dividing its weights by m. □

Under the identification $N_{Y/S}^{-1} \simeq F$, the local generator $dw_i|_Y$ corresponds to f_i. Thus the first obstruction class is

$$u_1(Y, S, L) = [\{-c_{ij}\, dw_i\}] = -\xi \quad \text{in } H^1(Y, N_{Y/S}^{-1}) \simeq H^1(B, F).$$

Koike's obstruction theorem applies to smooth weights [Koi21, Theorem 1.4]. Wu also computes the smooth obstruction in Grauert's example [Wu26, Example 3.7]. Lemma 2.2 gives the obstruction at C^1 regularity.

2.3. **Cusp correction and a global metric.** To construct the locally Lipschitz metric, we add a multiple of the norm of the defining section to a smooth local weight. Away from the curve this term supplies positive curvature in both the tangential and normal directions.

Lemma 2.4 (Cusp correction)**.** *Let D be a compact smooth connected curve in a complex surface X, with $D^2 < 0$, and let H be a line bundle near D whose restriction admits a smooth flat metric. There are smooth metrics h_0 on H and k on $\mathcal{O}_X(D)$, a neighborhood U of D, and $\Lambda > 0$ such that*

$$h_{\mathrm{loc}} = h_0 e^{-\Lambda s}, \qquad s = \|s_D\|_k,$$

has locally Lipschitz psh weights on U and smooth strictly psh weights on $U \setminus D$.

Proof. Extend the flat metric on $H|_D$ to a smooth metric h_0 near D and put $\theta = \Theta_{h_0}(H)$. Then $\theta|_{TD} = 0$. Since $\deg N_{D/X} = D^2 < 0$, choose a metric on $N_{D/X}$ with strictly negative curvature and extend it to a smooth metric k on $\mathcal{O}_X(D)$. Indeed, if γ is an initial curvature and ω_D is an area form of integral one, the equation

$$dd^c v = D^2\omega_D - \gamma, \qquad \int_D (D^2\omega_D - \gamma) = 0$$

is solvable on the compact curve. Multiplication by e^{-v} gives the prescribed negative curvature. Smooth extensions follow by extending the logarithm of the ratio of two metrics. Put

$$\eta = -\Theta_k(\mathcal{O}_X(D)), \qquad q = \|s_D\|_k^2, \qquad s = q^{1/2}.$$

In coordinates $D = \{w = 0\}$, with $s_D = wd$ and $\|d\|_k^2 = e^\beta$, one has

$$q = |w|^2 e^\beta, \qquad s = |w| e^{\beta/2}, \qquad \eta = dd^c\beta, \qquad \eta|_{TD} > 0. \tag{2.9}$$

Shrink finitely many relatively compact charts covering D until $|w\beta_w| \leq 1/2$. Define smooth vector fields

$$H_1 = \partial_z - \frac{w\beta_z}{1 + w\beta_w}\partial_w, \qquad H_2 = \partial_w.$$

They form a frame, with $H_1|_D = \partial_z$, and on $w \neq 0$

$$\partial \log q(H_1) = 0, \qquad \partial \log q(H_2) = \frac{1}{w} + \beta_w.$$

The tensor identity

$$dd^c s = \frac{s}{2}\eta + \frac{s}{4}\frac{i}{2\pi}\partial \log q \wedge \bar\partial \log q \tag{2.10}$$

therefore gives, in the frame (H_1, H_2),

$$(dd^c s)_{1\bar{1}} = \tfrac{s}{2}\eta_{1\bar{1}}, \qquad (dd^c s)_{1\bar{2}} = \tfrac{s}{2}\eta_{1\bar{2}},$$
$$(dd^c s)_{2\bar{2}} = \tfrac{s}{2}\eta_{2\bar{2}} + \tfrac{s}{4}|w^{-1} + \beta_w|^2.$$

These coefficients are tensor evaluations in a smooth frame, with the common factor $i/(2\pi)$ omitted. Choose uniform constants $\mu > 0$, $A \geq 1$ with

$$\eta_{1\bar{1}} \geq \mu, \qquad |\eta_{1\bar{2}}| + |\eta_{2\bar{2}}| \leq A, \qquad A^{-1} \leq e^{\beta} \leq A.$$

Since $|1 + w\beta_w| \geq 1/2$,

$$(dd^c s)_{2\bar{2}} \geq -\frac{A}{2}s + \frac{s}{16|w|^2} \geq -\frac{A}{2}s + \frac{1}{16As} \geq \frac{1}{32As}$$

after imposing $s^2 \leq 1/(16A^2)$. Smoothness and $\theta_{1\bar{1}}|_D = 0$ also give $|\theta_{1\bar{1}}| \leq C|w| \leq CA^{1/2}s$. Consequently there are $a, b, K > 0$, independent of the chart, such that

$$\begin{aligned} (dd^c s)_{1\bar{1}} \geq as, \qquad (dd^c s)_{2\bar{2}} \geq b/s, \quad |(dd^c s)_{1\bar{2}}| \leq Ks, \\ \theta_{1\bar{1}} \geq -Ks, \qquad \theta_{2\bar{2}} \geq -K, \qquad |\theta_{1\bar{2}}| \leq K. \end{aligned} \tag{2.11}$$

First fix Λ so that

$$\Lambda a - K > 0, \qquad (\Lambda a - K)\frac{\Lambda b}{2} > (K+1)^2. \tag{2.12}$$

Then shrink U until $Ks \leq \Lambda b/2$ and $\Lambda K s \leq 1$. For $T = \theta + \Lambda dd^c s$,

$$T_{1\bar{1}} \geq (\Lambda a - K)s > 0, \qquad T_{2\bar{2}} \geq \frac{\Lambda b}{2s} > 0, \qquad |T_{1\bar{2}}| \leq K + 1,$$

and hence

$$\det T \geq (\Lambda a - K)\frac{\Lambda b}{2} - (K+1)^2 > 0.$$

Thus T is strictly positive on $U \setminus D$. If φ_0 is a local weight of h_0, the weight $\varphi_0 + \Lambda s$ is continuous across D and psh off D, so the removable-singularity theorem makes it psh on the whole chart. Finally, $s = |w|e^{\beta/2}$ is locally Lipschitz, proving the assertion. □

Proposition 2.5 (Global cusp metric)**.** *Let X be a smooth projective surface, $D \subset X$ a smooth connected curve with $D^2 < 0$, and H a line bundle such that $H|_D$ admits a smooth flat metric. Suppose*

$$A = H^{\otimes m} \otimes \mathcal{O}_X(-a_0 D)$$

is ample for some integers $m, a_0 > 0$. Then H admits a locally Lipschitz semipositive Hermitian metric, smooth with strictly positive curvature on $X \setminus D$.

Proof. Extend the smooth metrics h_0 and k from Lemma 2.4 to X, and set $q = \|s_D\|_k^2$ and $s = q^{1/2}$ globally. Write $\theta = \Theta_{h_0}(H)$ and $u_{\mathrm{loc}} = \Lambda s$. Let h_A be a smooth metric with $\omega_A = \Theta_{h_A}(A) > 0$. The identity of line bundles $H^{\otimes m} = A \otimes \mathcal{O}_X(a_0 D)$ defines a global smooth function G by

$$h_A \otimes k^{a_0} = h_0^m e^{-G}.$$

For the canonical section s_D of $\mathcal{O}_X(D)$, the Poincaré–Lelong formula gives

$$dd^c \log \|s_D\|_k^2 = [D] - \Theta_k(\mathcal{O}_X(D)). \tag{2.13}$$

Hence the divisor metric $h_{[D]} = ke^{-\log q}$ has curvature $[D]$. Thus the auxiliary metric and its weight relative to h_0 are

$$h_*^m = h_A \otimes h_{[D]}^{a_0}, \qquad h_* = h_0 e^{-\psi}, \qquad \psi = \frac{a_0}{m}\log q + \frac{G}{m}, \tag{2.14}$$
$$\theta + dd^c\psi = \frac{1}{m}\omega_A + \frac{a_0}{m}[D].$$

In particular, ψ is strictly θ-psh off D and tends uniformly to $-\infty$ on approach to D.

Fix $\varepsilon > 0$. Choose an even nonnegative smooth function χ_ε supported in $(-\varepsilon, \varepsilon)$ with integral one, and put

$$\rho_\varepsilon(t) = \int_{\mathbb{R}} |t-a|\chi_\varepsilon(a)\,da, \qquad \mathcal{M}_\varepsilon(x,y) = \frac{x+y+\rho_\varepsilon(x-y)}{2}.$$

Then $\rho_\varepsilon'' \geq 0$, $|\rho_\varepsilon'| \leq 1$, and $\rho_\varepsilon(t) = |t|$ for $|t| \geq \varepsilon$. Consequently $\mathcal{M}_\varepsilon$ is convex, nondecreasing in both variables, and

$$\mathcal{M}_\varepsilon(x+c, y+c) = c + \mathcal{M}_\varepsilon(x,y), \qquad \mathcal{M}_\varepsilon(x,y) = \max\{x,y\} \quad (|x-y| \geq \varepsilon).$$

Writing $\mathcal{L}_f(v) = \sum_{j,k} f_{z_j\bar{z}_k} v_j \overline{v_k}$ for the Levi form of a C^2 function f, we obtain for smooth functions p_1, p_2

$$\mathcal{L}_{\mathcal{M}_\varepsilon(p_1,p_2)}(v) = \sum_{j=1}^{2} (\mathcal{M}_\varepsilon)_j \mathcal{L}_{p_j}(v) + \sum_{j,k=1}^{2} (\mathcal{M}_\varepsilon)_{jk} \partial p_j(v) \overline{\partial p_k(v)}.$$

The second term is nonnegative, and the coefficients in the first term are nonnegative and sum to one. Thus strict plurisubharmonicity is preserved.

Choose $b > 0$ so that $\{q \leq 2b\} \subset U$, where the local metric is defined, and then choose $N \in \mathbb{R}$ and $0 < a < b/4$ such that

$$\begin{aligned} \psi + N &> u_{\mathrm{loc}} + 2\varepsilon \quad \text{on } \{b/2 \leq q \leq b\}, \\ \psi + N &< u_{\mathrm{loc}} - 2\varepsilon \quad \text{on } \{0 < q \leq 2a\}. \end{aligned} \tag{2.15}$$

First take N large on the compact shell to obtain the first inequality, then take a small enough that

$$\frac{a_0}{m}\log(2a) + \max_X (G/m) + N < -2\varepsilon.$$

Define

$$h = h_0 e^{-u}, \qquad u = \begin{cases} u_{\mathrm{loc}}, & q \leq a, \\ \mathcal{M}_\varepsilon(u_{\mathrm{loc}}, \psi + N), & a < q < b, \\ \psi + N, & q \geq b. \end{cases} \tag{2.16}$$

The expressions coincide on collars of $q = a$ and $q = b$ by (2.15). On the annulus, a local weight is

$$\varphi_0 + u = \mathcal{M}_\varepsilon(\varphi_0 + u_{\mathrm{loc}}, \varphi_0 + \psi + N),$$

so it is smooth and strictly psh. The inner weight is psh by Lemma 2.4, and the outer one is strictly psh by (2.14). The metric is locally Lipschitz near D and smooth with strictly positive curvature elsewhere. □

Proposition 2.6. *For the ruled surface (2.2), L admits a locally Lipschitz semipositive Hermitian metric, smooth with strictly positive curvature on $S \setminus Y$. Its curvature has bounded local potentials and minimal singularities.*

Proof. Equation (2.4) and Proposition 2.1 give

$$Y^2 = -d < 0, \qquad L|_Y \simeq \mathcal{O}_Y, \qquad L^{\otimes 2} \otimes \mathcal{O}_S(-Y) \text{ ample.}$$

Apply Proposition 2.5 with $(X, D, H, m, a_0) = (S, Y, L, 2, 1)$. Write the resulting curvature as $T = \theta + dd^c u \geq 0$, where u is continuous on compact S. For every global quasi-psh function $v \not\equiv -\infty$ with $\theta + dd^c v \geq 0$,

$$v \leq u + \sup_S v - \inf_S u.$$

This is the defining comparison for minimal singularities, and the local potentials $\varphi_0 + u$ are bounded. □

Remark 2.7. The choices $g = 2$, $F = \mathcal{O}_B(p)$ and $F = K_B$ yield respectively

$$(Y^2, L^2) = (-1, 1), \qquad (Y^2, L^2) = (-2, 2).$$

Proposition 2.6 gives bounded potentials with minimal singularities for both classes considered in [Zha25, Section 4]. Proposition 2.3 excludes C^1 semipositive weights in either case.

3. The regularity criterion

For a semipositive Hermitian metric on A, the squared norm on the dual bundle gives a psh exhaustion. To recover a metric from an exhaustion, we first average over rotations of the fibers. A suitable sublevel set then meets each fiber in a disc whose radius determines the metric. We carry out both constructions for the local regularity classes

$$\begin{aligned} \mathcal{R} \in \{C^r : r \in \mathbb{Z}_{\geq 0} \cup \{\infty\}\} \cup \{C^\omega\} \\ \cup \{C^{k,\alpha}_{\mathrm{loc}} : k \in \mathbb{Z}_{\geq 0},\ 0 < \alpha \leq 1\}. \end{aligned} \tag{3.1}$$

For every class in (3.1), the positive metric coefficients h_i, their reciprocals, and the weights $-\log h_i$ have the same regularity.

Lemma 3.1. *Let $A \to Z$ be a holomorphic line bundle over a compact complex manifold. If h is a semipositive Hermitian metric of class $\mathcal{R}$, then $\rho(v) = \|v\|^2_{h^{-1}}$ is a psh exhaustion of $\mathrm{Tot}(A^{-1})$ of the same class.*

Proof. Let e_i be a local frame of A over $B_i \subset Z$, with $\|e_i\|^2_h = e^{-\varphi_i}$. For $v = \lambda_i e_i^*$,

$$\rho(z, \lambda_i) = |\lambda_i|^2 e^{\varphi_i(z)}. \tag{3.2}$$

This formula gives the claimed regularity on the total space. Off the zero section, $\log \rho = \log |\lambda_i|^2 + \varphi_i$ is psh, so ρ is psh there by convex increasing composition. Continuity and removability across the zero section give plurisubharmonicity everywhere.

Fix a smooth Hermitian metric k on A^{-1}. The ratio of h^{-1} to k is a positive continuous function on the compact base. Its minimum $m > 0$ gives $\rho(v) \geq m\|v\|^2_k$, so each sublevel of ρ is a closed subset of a compact disc bundle. □

Lemma 3.2 (Compact-group averaging)**.** *If a compact group G acts continuously on a locally compact Hausdorff space and F_0 is a continuous exhaustion, then $F(x) = \int_G F_0(gx)\, dg$, with normalized Haar measure dg, is an exhaustion. For the fiberwise circle action on a holomorphic line bundle, averaging preserves plurisubharmonicity and every class in (3.1).*

Proof. Continuity follows from uniform continuity on compact orbit sets. For $K_c = \{F_0 \leq c\}$,

$$\{F \leq c\} \subset G \cdot K_c.$$

Indeed, $x \notin G \cdot K_c$ implies $\min_{g \in G} F_0(gx) > c$, and therefore $F(x) > c$. The set $G \cdot K_c$ is the continuous image of the compact set $G \times K_c$, so its closed subset $\{F \leq c\}$ is compact.

For the circle action $R_\theta(v) = e^{i\theta}v$, put

$$F(v) = \frac{1}{2\pi}\int_0^{2\pi} F_0(R_\theta v)\, d\theta. \tag{3.3}$$

If F_0 is psh, integration of the submean inequality on each holomorphic disc gives the same inequality for F. Invariance under rotation follows from translation invariance of $d\theta$.

In real bundle coordinates $x = (z, \lambda)$, write $T_\theta x = (z, e^{i\theta}\lambda)$. The maps T_θ are linear with uniformly bounded norms. For $F_0 \in C^r$, $r \geq 1$, differentiation on compact coordinate sets yields

$$D^\beta F(x) = \frac{1}{2\pi}\int_0^{2\pi} D_x^\beta (F_0 \circ T_\theta)(x)\, d\theta, \qquad |\beta| \leq r.$$

For $F_0 \in C^{k,\alpha}_{\mathrm{loc}}$, a finite cover of the compact orbit set gives, uniformly in θ,

$$|D^\beta(F_0 \circ T_\theta)(x) - D^\beta(F_0 \circ T_\theta)(y)| \leq C|x-y|^\alpha, \qquad |\beta| = k.$$

Integration gives the same estimate for $D^\beta F$.

For the analytic case, $\mathcal{G}(x,\theta) = F_0(T_\theta x)$ is jointly real analytic near the compact set $\{x_0\} \times S^1$. A finite cover by analytic coordinate neighborhoods and Cauchy's estimates give constants $M, R > 0$ such that

$$\left| \frac{D_x^\beta \mathcal{G}(x_0,\theta)}{\beta!} \right| \leq M R^{-|\beta|} \qquad (\beta \in \mathbb{Z}^m_{\geq 0},\ \theta \in S^1), \tag{3.4}$$

where m is the real coordinate dimension. The Taylor series in x converges uniformly in θ on every smaller polydisc. Integrating term by term gives

$$F(x) = \sum_\beta \left(\frac{1}{2\pi}\int_0^{2\pi} \frac{D_x^\beta \mathcal{G}(x_0,\theta)}{\beta!}\, d\theta \right) (x - x_0)^\beta.$$

This convergent power series proves $F \in C^\omega$. □

Proposition 3.3 (Recovery of the metric)**.** *Let $A \to Z$ be a holomorphic line bundle over a compact complex manifold. If $\mathrm{Tot}(A^{-1})$ admits a psh exhaustion of class $\mathcal{R}$ in (3.1), then A admits a semipositive Hermitian metric of the same class.*

Proof. Let F be the circle average (3.3) of the given exhaustion. For a local frame e_i of A over B_i, define

$$U_i(z,t) = F(e^{t/2}e_i^*(z)), \qquad V_i(z,\zeta) = F(e^{\zeta/2}e_i^*(z)).$$

Both parametrizations are real analytic, so U_i and V_i inherit the class $\mathcal{R}$. The second parametrization is holomorphic, so V_i is psh. Circle invariance gives

$$V_i(z,\zeta) = U_i(z, \mathrm{Re}\,\zeta), \qquad F(\lambda_i e_i^*(z)) = U_i(z, \log|\lambda_i|^2) \quad (\lambda_i \neq 0).$$

Fix z and write $u(t) = U_i(z,t)$. Since V_i is independent of $\mathrm{Im}\,\zeta$, its subharmonicity in ζ gives $u'' \geq 0$ in the sense of distributions. Thus u is convex. Continuity at the zero section and properness of F give

$$\lim_{t\to-\infty} u(t) = F(0_z), \qquad \lim_{t\to+\infty} u(t) = +\infty. \tag{3.5}$$

If $u(b) < u(a)$ for $a < b$, the negative secant slope $d = (u(b) - u(a))/(b-a)$ would give

$$\frac{u(a) - u(x)}{a - x} \leq d < 0, \qquad u(x) \geq u(a) - d(a-x) \longrightarrow +\infty \quad (x \to -\infty),$$

contrary to (3.5). Thus u is nondecreasing.

Choose $c > \max_{z\in Z} F(0_z)$. The intermediate value theorem gives a point s with $u(s) = c$. If $u(s) = u(t) = c$ for $s < t$, then for $x < s$ the ordered secant slopes give

$$0 \le \frac{u(s) - u(x)}{s - x} \le \frac{u(t) - u(s)}{t - s} = 0.$$

This would give $u(x) = c$ for all $x < s$, again contradicting (3.5). Consequently there is a unique $\sigma_i(z)$ such that

$$U_i(z, \sigma_i(z)) = c, \qquad U_i(z, t) \le c \iff t \le \sigma_i(z). \tag{3.6}$$

In particular,

$$\{F < c\} \cap A^{-1}|_{B_i} = \{(z, \lambda_i) : |\lambda_i|^2 < e^{\sigma_i(z)}\}. \tag{3.7}$$

For $s_0 = \sigma_i(z_0)$ and $\varepsilon > 0$,

$$U_i(z_0, s_0 - \varepsilon) < c < U_i(z_0, s_0 + \varepsilon).$$

These strict inequalities persist near z_0 and force $s_0 - \varepsilon < \sigma_i(z) < s_0 + \varepsilon$. Thus σ_i is continuous.

To prove that $-\sigma_i$ is psh, let f be a holomorphic map from a neighborhood of $\overline{\Delta}$ to B_i. Let H be the harmonic extension of $\sigma_i \circ f|_{\partial\Delta}$, continuous on $\overline{\Delta}$. Since Δ is simply connected, there is a holomorphic g with $\operatorname{Re} g = H$. The function

$$q(\zeta) = F\big(e^{g(\zeta)/2} e_i^*(f(\zeta))\big) = U_i(f(\zeta), H(\zeta))$$

is subharmonic on Δ, and the second expression extends continuously to the boundary with value c. The maximum principle and (3.6) give

$$q \le c \quad\Longrightarrow\quad H \le \sigma_i \circ f \quad\Longrightarrow\quad -\sigma_i(f(0)) \le \frac{1}{2\pi} \int_0^{2\pi} -\sigma_i(f(e^{i\theta}))\, d\theta.$$

The submean inequality for every such disc proves that $-\sigma_i$ is psh.

If $e_j = g_{ij} e_i$, then $\lambda_j = g_{ij}\lambda_i$, and the radii of the same sublevel in (3.7) satisfy

$$e^{\sigma_j} = |g_{ij}|^2 e^{\sigma_i}, \qquad \sigma_j = \sigma_i + \log |g_{ij}|^2.$$

Thus

$$\psi_i = -\sigma_i, \qquad \psi_j = \psi_i - \log |g_{ij}|^2, \qquad \|e_i\|_h^2 = e^{-\psi_i} = e^{\sigma_i} \tag{3.8}$$

defines a continuous Hermitian metric on A with $\Theta_h(A) = dd^c\psi_i \ge 0$.

We now recover the regularity of F. For C^1 data, convexity gives, for any $a < \sigma_i(z)$,

$$U_{i,t}(z, \sigma_i(z)) \ge \frac{c - U_i(z, a)}{\sigma_i(z) - a} > 0. \tag{3.9}$$

The implicit function theorem gives $\sigma_i \in C^r$ whenever $F \in C^r$, for $1 \le r \le \infty$. Its real analytic version gives the same conclusion for C^ω.

For $C^{0,\alpha}_{\mathrm{loc}}$ data, omit the index i and choose

$$a < b < \sigma(z_0) < d, \qquad U(z_0, a) < U(z_0, b).$$

The limits (3.5) permit this choice. By continuity, there are a relatively compact base ball B around z_0 and $m > 0$ such that

$$b < \sigma(z) < d, \qquad \frac{U(z, b) - U(z, a)}{b - a} \ge m \qquad (z \in \overline{B}).$$

The local Hölder regularity of U on a neighborhood of $\overline{B} \times [b, d]$ gives one constant C such that

$$|U(z, t) - U(w, t)| \le C|z - w|^\alpha \qquad (z, w \in B,\ t \in [b, d]).$$

Ordered secant slopes then yield

$$U(z,t) - U(z,s) \geq m(t-s), \qquad b \leq s \leq t \leq d.$$

If $\sigma(z) \geq \sigma(w)$, this implies

$$\begin{aligned} m|\sigma(z) - \sigma(w)| &\leq U(z,\sigma(z)) - U(z,\sigma(w)) \\ &= U(w,\sigma(w)) - U(z,\sigma(w)) \leq C|z-w|^{\alpha}. \end{aligned}$$

Interchanging z, w gives the same bound in the other case. Hence

$$|\sigma(z) - \sigma(w)| \leq (C/m)|z-w|^{\alpha}, \qquad 0 < \alpha \leq 1. \tag{3.10}$$

Finally, suppose $U \in C^{k,\alpha}_{\mathrm{loc}}$ with $k \geq 1$. The implicit function theorem first gives $\sigma \in C^k$. On a smaller convex coordinate neighborhood, $x \mapsto (x, \sigma(x))$ is Lipschitz and $U_t(x,\sigma(x))$ is bounded below by a positive constant. Differentiating the level equation gives

$$\sigma_{x_j} = -\frac{U_{x_j}(x,\sigma(x))}{U_t(x,\sigma(x))}, \qquad U_t(x,\sigma(x)) D^{\beta}\sigma = -P_{\beta}, \quad |\beta| = k,$$

where P_β is a finite sum of products of derivatives of U of order at most k, evaluated on the graph, and derivatives of σ of order at most $k-1$. For $k=1$, the first formula has α-Hölder numerator and denominator. For $k \geq 2$,

$$D^j\sigma \in C^{0,1}_{\mathrm{loc}} \ (j<k), \qquad D^jU \in C^{0,1}_{\mathrm{loc}} \ (j<k), \qquad D^kU \in C^{0,\alpha}_{\mathrm{loc}}.$$

Composition with the Lipschitz graph and multiplication preserve these local Hölder estimates. Division by U_t therefore gives $D^\beta\sigma \in C^{0,\alpha}_{\mathrm{loc}}$. Formula (3.8) now proves the assertion in every class. □

Theorem 3.4 (Regularity criterion)**.** *Let $A \to Z$ be a holomorphic line bundle over a compact complex manifold. For each class $\mathcal{R}$ in (3.1), the following are equivalent.*

(i) $\mathrm{Tot}(A^{-1})$ *admits a psh exhaustion of class $\mathcal{R}$.*
(ii) *A admits a semipositive Hermitian metric of class $\mathcal{R}$.*

Proof. Proposition 3.3 gives (i)⇒(ii). Lemma 3.1 gives (ii)⇒(i). □

For C^2 data, the relation between the two Levi forms is explicit. Differentiating $U(z,\sigma(z)) = c$ twice and using $V(z,\zeta) = U(z, \mathrm{Re}\,\zeta)$ gives

$$\mathcal{L}_{-\sigma}(\xi) = \frac{\mathcal{L}_V(\xi, 2\partial\sigma(\xi))}{U_t} \geq 0. \tag{3.11}$$

The identity is evaluated along the graph $t = \sigma(z)$. For the dual norm, direct differentiation and completion of the square give

$$\mathcal{L}_{|\lambda|^2 e^{\varphi}}(\xi,\eta) = e^{\varphi}\left(|\eta + \lambda\partial\varphi(\xi)|^2 + |\lambda|^2 \mathcal{L}_{\varphi}(\xi)\right) \geq 0.$$

Proof of Theorem 1.1. Take the ruled surface S, the section Y, and the line bundle L constructed in Section 2, for example with $F = K_B$. The total space $M = \mathrm{Tot}(L^{-1})$ has the compactification

$$M = \mathbf{P}(\mathcal{O}_S \oplus L^{-1}) \setminus \mathbf{P}(L^{-1}), \qquad v \longmapsto [(1,v)]. \tag{3.12}$$

The inverse sends $[(a,b)]$, with $a \neq 0$, to b/a. Since S is projective and L is algebraic, the projective bundle in (3.12) is projective. Therefore $\dim_{\mathbb{C}} M = 3$, M is smooth and quasi-projective, and the restriction of a projective Kähler form makes it Kähler.

Let h be the locally Lipschitz semipositive metric of Proposition 2.6. Lemma 3.1 gives the locally Lipschitz psh exhaustion

$$\rho(v) = \|v\|^2_{h^{-1}}. \tag{3.13}$$

If M admitted a C^1 psh exhaustion, Theorem 3.4 would give a C^1 semipositive Hermitian metric on L, contrary to Proposition 2.3. □

4. A non-Kähler modification over one point

We first show that a modification with a compact center preserves the existence of a C^r psh exhaustion. We then place Hironaka's opposite-order blowup construction inside the exceptional plane over a point to prove Theorem 1.2.

Proposition 4.1. *Let $f : X' \to X$ be a proper surjective holomorphic map of complex manifolds, and suppose that $f : X' \setminus f^{-1}(K) \to X \setminus K$ is a biholomorphism for some compact $K \subset X$. For $r \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, X' admits a C^r psh exhaustion if and only if X does.*

Proof. If u is a C^r psh exhaustion of X, then $u \circ f$ is C^r and psh, and

$$\{u \circ f \leq a\} = f^{-1}(\{u \leq a\})$$

is compact by properness.

Conversely, let F be a C^r psh exhaustion of X'. Choose a relatively compact open neighborhood U of K and take $c > \sup_{f^{-1}(\overline{U})} F$. Let χ be a smooth convex nondecreasing function, zero on $(-\infty, c]$ and tending to $+\infty$ as $t \to +\infty$. Let $j : X \setminus K \to X' \setminus f^{-1}(K)$ be the holomorphic inverse of f. Then

$$G = \chi \circ F \circ j \quad \text{on } X \setminus K, \qquad G = 0 \quad \text{on } K$$

vanishes throughout U, because $F < c$ on $f^{-1}(U)$. It is therefore C^r across K. Convex nondecreasing composition gives plurisubharmonicity on $X \setminus K$, and G is constant on U. The identity $G \circ f = \chi \circ F$ gives

$$\{G \leq a\} = f(\{\chi \circ F \leq a\}).$$

The right-hand side is compact, since $\chi \circ F$ is an exhaustion. □

Proposition 4.2. *For every point x of a connected smooth complex threefold X, there is a connected smooth non-Kähler threefold N and a proper modification $\mu : N \to X$ that is biholomorphic over $X \setminus \{x\}$.*

Proof. Following the construction in [Bar15, §6], first blow up x,

$$b : W = \mathrm{Bl}_x X \to X, \qquad E = b^{-1}(x) \simeq \mathbf{P}^2.$$

In homogeneous coordinates $[u : v : w]$ on E, the line and conic

$$A = \{w = 0\}, \qquad B = \{uv = w^2\}$$

meet transversely in E at $p = [1 : 0 : 0]$ and $q = [0 : 1 : 0]$. Set $W_p = W \setminus \{q\}$ and $W_q = W \setminus \{p\}$. On W_p and W_q respectively, form

$$N_p = \mathrm{Bl}_{\widetilde{B \cap W_p}}(\mathrm{Bl}_{A \cap W_p} W_p),$$
$$N_q = \mathrm{Bl}_{\widetilde{A \cap W_q}}(\mathrm{Bl}_{B \cap W_q} W_q).$$

Here a tilde denotes strict transform. Over $W_p \cap W_q = W \setminus \{p, q\}$ the two centers are disjoint. The two blowup orders have a canonical isomorphism there, agreeing with the identity off $A \cup B$. Glue by this isomorphism to obtain N, with maps

$$\nu : N \longrightarrow W, \qquad \mu = b \circ \nu : N \longrightarrow X.$$

The centers are smooth, so N is smooth. Points with distinct images in W can be separated using neighborhoods in W, while points over the same image lie together in N_p or N_q. Thus N is Hausdorff. Properness is local on the base, so ν and μ are proper. The two connected blowup

charts overlap, making N connected. Every center lies in $E = b^{-1}(x)$, so μ is biholomorphic over $X \setminus \{x\}$.

The obstruction to a Kähler form is a homology relation between two exceptional curves, as in [Sta, Tag 08J1]. Let D_A, D_B be the exceptional surfaces dominating A, B, with the induced maps

$$f_A : D_A \longrightarrow A, \qquad f_B : D_B \longrightarrow B.$$

Locally at a crossing they are ruled surfaces or blowups of ruled surfaces at a point. They are therefore smooth, and properness of ν makes them compact. Let $\alpha, \beta \in H_2(N, \mathbb{R})$ be their general fiber classes. Since A and B are rational, their point divisors are linearly equivalent. Pulling back shows that all fibers of f_A represent α, and all fibers of f_B represent β.

Near p, choose coordinates (z_1, z_2, z_3) with $A = \{z_2 = z_3 = 0\}$ and $B = \{z_1 = z_3 = 0\}$. The first blowup has the chart

$$(z_1, z_2, t) \longmapsto (z_1, z_2, z_3 = z_2 t).$$

Its exceptional divisor is $z_2 = 0$, and the strict transform of B is $z_1 = t = 0$. A chart of the second blowup is

$$(r, z_2, t) \longmapsto (z_1 = rt, z_2, t).$$

On the strict transform $D_A = \{z_2 = 0\}$, the map f_A is $(r, t) \mapsto z_1 = rt$. Its fiber over p is therefore

$$\operatorname{div}(rt) = \{r = 0\} + \{t = 0\} = R_p + T_p.$$

The closures of these two components are respectively the strict transform of the first exceptional fiber and the exceptional curve of its surface blowup. Both are smooth rational curves, and the factorization rt gives multiplicity one. The second exceptional surface D_B has fiber T_p over p. Interchanging A and B in the same calculation at q gives

$$\alpha = [R_p] + [T_p], \quad \beta = [T_p], \qquad \beta = [R_q] + [T_q], \quad \alpha = [T_q].$$

Subtracting each pair gives $[R_p] = \alpha - \beta$ and $[R_q] = \beta - \alpha$, hence

$$[R_p] + [R_q] = 0 \quad \text{in } H_2(N, \mathbb{R}). \tag{4.1}$$

Integration of a Kähler form over this homologically trivial effective cycle would give

$$0 = \int_{R_p} \omega + \int_{R_q} \omega > 0,$$

a contradiction. □

Proof of Theorem 1.2. Apply Proposition 4.2 to M at x, and let ρ be the $C^{0,1}_{\mathrm{loc}}$ psh exhaustion from Theorem 1.1. Since μ is smooth and proper, $\rho \circ \mu$ is a $C^{0,1}_{\mathrm{loc}}$ psh exhaustion. A $C^{1,0}$ psh exhaustion on N would descend after truncation by Proposition 4.1, contradicting Theorem 1.1. □

5. Surface regularization and boundary criteria

Throughout this section, $X = S \setminus \operatorname{supp} D$, where S is a smooth compact connected complex surface and $D \neq 0$ is effective. We identify reduced divisors with their supports. A *divergent psh collar* along a boundary component E is a psh function on $U \setminus E$, for a neighborhood U of E, tending uniformly to $+\infty$ as one approaches E. We prove regularization by constructing smooth collars and extending them to an exhaustion of X.

5.1. **Boundary tools.** The intersection matrix of a boundary component separates three cases. Negative definiteness obstructs an exhaustion, a positive eigenvalue gives a strictly psh collar, and a zero largest eigenvalue gives a divisor of square zero.

Lemma 5.1. (i) *If a connected component of* $\operatorname{supp} D$ *has negative definite intersection matrix,* X *has no continuous psh exhaustion.*

(ii) *Smooth divergent psh collars at all connected boundary components give a smooth psh exhaustion of* X*.*

(iii) *For a connected reduced divisor* $E = \bigcup E_i$ *with simple normal crossings (SNC), let* λ *be the largest eigenvalue of* $I = (E_i \cdot E_j)$*. If* $\lambda > 0$*,* E *has a smooth strictly psh divergent collar. If* $\lambda = 0$*,* $I \leq 0$*,* $\dim \ker I = 1$*, and*

$$A = \sum_i a_i E_i, \qquad a_i \in \mathbb{Z}_{>0}, \qquad A \cdot E_i = A^2 = 0. \tag{5.1}$$

When S *is projective,* A *is nef.*

Proof. For (i), let E be the boundary component. Grauert's theorem [Gra62, §4, subsection 8(e)] contracts E to a point p of a normal surface Z. Write the contraction as $q : W \to Z$, with $W \backslash E \to Z \backslash \{p\}$ a biholomorphism. An exhaustion ρ on X would therefore give $u = \rho \circ (q|_{W\backslash E})^{-1}$, psh and divergent at p. Choose a finite local projection

$$\pi : V \longrightarrow \Delta^2, \qquad \pi^{-1}(0) = \{p\}, \qquad v(z) = \max_{y \in \pi^{-1}(z)} u(y).$$

The function v is continuous on $\Delta^2 \setminus \{0\}$. Off the branch locus it is a finite maximum of psh functions. Properness of π gives local boundedness at each remaining point away from 0, so removability makes v psh there as well. Fix $0 < r < 1$ and set

$$M_r = \max_{\{|z_1| \leq r,\ |z_2| = r\}} v < \infty.$$

For $0 < |z_1| < r$, the subharmonic maximum principle on the vertical disc gives $v(z_1, z_2) \leq M_r$ for $|z_2| < r$. Continuity gives the same bound when $z_1 = 0$ and $z_2 \neq 0$. Hence $u \leq M_r$ near p, contradicting its divergence.

For (ii), choose disjoint neighborhoods W_E of the boundary components so that each collar Φ_E is defined on a neighborhood of $\overline{W}_E \setminus E$. Choose $T_E > \sup_{\partial W_E} \Phi_E$ and a smooth convex nondecreasing χ_E, zero on $(-\infty, T_E]$ and tending to $+\infty$. Define

$$\rho_E(x) = \begin{cases} \chi_E(\Phi_E(x)), & x \in W_E \setminus E, \\ 0, & x \notin W_E. \end{cases}$$

The two formulas agree on a neighborhood of ∂W_E, and

$$\mathcal{L}_{\chi_E \circ \Phi_E}(\xi) = \chi_E'(\Phi_E)\mathcal{L}_{\Phi_E}(\xi) + \chi_E''(\Phi_E)|\partial \Phi_E(\xi)|^2 \geq 0.$$

Thus $\rho = \sum_E \rho_E$ is smooth and psh. Every sequence leaving compact subsets of X approaches D along a subsequence, where the corresponding summand tends to infinity. This proves properness.

For (iii), Perron–Frobenius applied to $I + c\mathrm{Id}$ shows that λ is simple and has a strictly positive eigenvector v. When $\lambda > 0$, both $v > 0$ and $Iv > 0$ persist under small perturbations. Approximate v by a rational vector and clear denominators to obtain $A = \sum a_i E_i$ with $a_i \in \mathbb{Z}_{>0}$ and $A \cdot E_i > 0$ for all i. Thus $\mathcal{O}_S(A)|_E$ is ample. A projective embedding by a power gives a Fubini–Study metric on this restriction. Extend its weights locally to S and patch them relative to a reference metric. The extensions have positive tangential curvature along E. Near a node, lift the embedding sections from E to a coordinate neighborhood in S. Their ratios

have differentials spanning the two-dimensional Zariski cotangent space of the node, and hence the ambient cotangent space. Their Fubini–Study weights have positive curvature in both directions. Keep these weights on smaller node neighborhoods when patching. The other extensions agree along E, so the patched metric retains positive tangential curvature at its smooth points.

Let $q = \|s_E\|^2$ for a smooth metric on $\mathcal{O}_S(E)$. At a smooth point of $E = \{w = 0\}$, write $q = |w|^2 e^{-\gamma}$. Then

$$\partial\bar{\partial} q|_E = e^{-\gamma}|_E \, dw \wedge d\bar{w}.$$

Away from the node neighborhoods the curvature matrix of the patched metric has the form $\left(\begin{smallmatrix} a & b \\ \bar{b} & d \end{smallmatrix}\right)$ in tangential and normal directions, with $a \geq \varepsilon > 0$. Adding Cq to its weights replaces d by $d + Ce^{-\gamma}$ along E. By compactness, one constant C makes $a(d + Ce^{-\gamma}) > |b|^2$. At the nodes $dd^c q = 0$, and at other points of E it is semipositive, so positivity is preserved there. Continuity now gives a metric h of positive curvature near all of E. On the complement,

$$dd^c(-\log \|s_A\|_h^2) = \Theta_h(\mathcal{O}_S(A)) > 0,$$

and $a_i > 0$ gives uniform divergence. For $\lambda = 0$, every other eigenvalue is negative. Since I has integer entries and rank one less than its size, its positive kernel has a rational generator, and clearing denominators proves (5.1). Effectiveness gives $A \cdot C \geq 0$ for every curve $C \not\subset E$, which proves nefness in the projective case. □

For a boundary resolution $p : \widetilde{S} \to S$, the restriction $\widetilde{S} \backslash p^{-1}(\operatorname{supp} D) \to X$ is a biholomorphism. Interior blowups preserve both exhaustion properties by Proposition 4.1.

Proposition 5.2. *A connected noncompact smooth complex surface X with a continuous psh exhaustion and $\mathcal{O}(X) \neq \mathbb{C}$ has a smooth psh exhaustion.*

Proof. Vîjîitu's theorem [Vîj22a, Theorem 1.1] gives holomorphic convexity. Let $r : X \to Y$ be the proper Remmert reduction and let τ be a smooth strictly psh exhaustion of the Stein space Y. Then

$$\rho = \tau \circ r, \qquad \{\rho \leq c\} = r^{-1}(\{\tau \leq c\}) \Subset X$$

is a smooth psh exhaustion. □

Proposition 5.3. *Let A be a line bundle on a compact Riemann surface B. The following are equivalent.*

(i) $\operatorname{Tot}(A^{-1})$ *admits a continuous psh exhaustion.*
(ii) $\deg A \geq 0$.
(iii) $\operatorname{Tot}(A^{-1})$ *admits a smooth psh exhaustion.*

Proof. For (i)⇒(ii), Theorem 3.4 gives a continuous semipositive metric on A. Its positive curvature current has total mass $\deg A$, so $\deg A \geq 0$.

For (ii)⇒(iii), put $d = \deg A \geq 0$. Choose an area form ω of integral one and a smooth reference metric h_0 of curvature θ_0. Since $d\omega - \theta_0$ has integral zero, the equation $dd^c f = d\omega - \theta_0$ has a smooth solution on B. The metric $h = e^{-f} h_0$ then satisfies

$$\Theta_h(A) = \theta_0 + dd^c f = d\omega \geq 0.$$

Lemma 3.1 gives the smooth psh exhaustion. The implication (iii)⇒(i) is immediate. □

5.2. **Current averaging and regularization.** The main step is to average currents supported on exhaustion levels. The resulting class is orthogonal to the boundary, and the Hodge index theorem relates it to a divisor of square zero. Theorem 1.3 follows from the next theorem and its quasi-projective corollary.

Theorem 5.4. *Let S be a compact Kähler surface and $D = \sum_i D_i \neq 0$ a reduced divisor. Suppose $X = S \setminus D$ admits a continuous psh exhaustion. Then it admits a smooth psh exhaustion. If X is not a modification of a Stein space, there is a divisor*

$$A = \sum_i a_i D_i, \qquad a_i \in \mathbb{Z}_{>0}, \qquad A \cdot D_i = A^2 = 0,$$

such that $\mathcal{O}_S(A)$ has a smooth Hermitian metric with zero curvature near D. Its canonical section gives a pluriharmonic divergent collar along D.

Lemma 5.5. *Let (S, ω) be a compact Kähler surface and $X \subset S$ a connected noncompact open set with a continuous psh exhaustion ρ. If X is not a modification of a Stein space, there is a positive $(1,1)$-current T on S such that*

$$dd^c T = 0, \qquad \int_S T \wedge \omega = 1, \qquad \operatorname{supp} T \Subset X, \qquad \nu(T, x) = 0 \quad (x \in S). \tag{5.2}$$

Proof. Let Σ_X^0 be the set of points at which every continuous psh exhaustion fails to be strongly psh. By [MS20, Lemma 2.1],

$$(t_0, \infty) \subset \rho(\Sigma_X^0) \tag{5.3}$$

for some t_0. For $t > t_0$, no smooth strictly psh function exists near $K_t = \{\rho = t\}$. Indeed, if v were such a function, properness permits $\delta > 0$ with the compact shell $\{t - \delta \leq \rho \leq t + \delta\}$ contained in its domain. After scaling and adding a constant, arrange $0 < v < 1$ there. For $0 < \eta < 1$, $0 < \varepsilon < \eta\delta/2$, set $g(s) = s + \eta|s - t|$. The function

$$\psi = \begin{cases} \max\{g(\rho), \rho + \varepsilon v\}, & |\rho - t| < \delta, \\ g(\rho), & |\rho - t| \geq \delta \end{cases}$$

is well defined because the formulas agree near $|\rho - t| = \delta$, where $g(\rho) - \rho = \eta|\rho - t| > \varepsilon v$. Both entries of the maximum are psh, since g is convex and increasing. On K_t the second entry strictly dominates, so $\psi = \rho + \varepsilon v$ is strongly psh near K_t. Finally $\psi \geq g(\rho)$ makes ψ an exhaustion. This contradicts $K_t \cap \Sigma_X^0 \neq \varnothing$.

Let $\mathcal{B}_t$ be the Banach space of continuous real $(1,1)$-forms on K_t, with the uniform norm induced by ω. Consider the subspace and open cone

$$V_t = \{dd^c f|_{K_t} : f \in C^\infty(S, \mathbb{R})\}, \qquad P_t = \{\eta \in \mathcal{B}_t : \eta > 0 \text{ on } K_t\}.$$

The absence of a smooth strictly psh function near K_t gives $V_t \cap P_t = \varnothing$. Hahn–Banach separation therefore gives a nonzero continuous functional ℓ_t with

$$\ell_t|_{V_t} = 0, \qquad \ell_t(\eta) \geq 0 \quad (\eta \geq 0).$$

The second assertion follows by applying the separating inequality to $\eta + \varepsilon\omega$ and letting $\varepsilon \downarrow 0$. Moreover $\ell_t(\omega) > 0$: otherwise $-C\omega \leq \eta \leq C\omega$ would force $\ell_t(\eta) = 0$ for every η. Define the current by

$$\langle T_t, \eta \rangle = \frac{\ell_t(\eta|_{K_t})}{\ell_t(\omega|_{K_t})}.$$

This defines a positive current supported on K_t, and $\ell_t|_{V_t} = 0$ gives $dd^c T_t = 0$. Hence

$$T_t \geq 0, \qquad dd^c T_t = 0, \qquad \operatorname{supp} T_t \subset K_t, \qquad \int_S T_t \wedge \omega = 1. \tag{5.4}$$

To average these currents, fix $t_0 < a < b$ and set

$$t_{n,j} = a + \frac{j - 1/2}{n}(b-a), \qquad T_n = \frac{1}{n}\sum_{j=1}^{n} T_{t_{n,j}}.$$

Each T_n has mass one and support in the compact shell $K = \rho^{-1}([a,b]) \Subset X$. A weakly convergent subsequence has a positive limit T satisfying

$$\int_S T \wedge \omega = 1, \qquad dd^c T = 0, \qquad \operatorname{supp} T \subset K.$$

Fix a coordinate ball $B(x, 2R) \Subset X$, and put $\beta = dd^c|z|^2$. Lelong–Jensen monotonicity for positive dd^c-closed currents [DNS22, Appendix B, (B.1)–(B.2)] gives, uniformly in t, for $0 < r < R$,

$$r^{-2}\int_{B(x,r)} T_t \wedge \beta \le R^{-2}\int_{B(x,R)} T_t \wedge \beta \le C_R. \tag{5.5}$$

Indeed, $\beta \le c_R\omega$ on $B(x,R)$, so one may take $C_R = c_R/R^2$, independently of t. Put $m_r = \inf_{B(x,r)} \rho$ and $M_r = \sup_{B(x,r)} \rho$. Since $\operatorname{supp} T_t \subset \{\rho = t\}$, only indices with $t_{n,j} \in [m_r, M_r]$ contribute. Their number $N_n(r)$ satisfies

$$N_n(r) \le \frac{n(M_r - m_r)}{b-a} + 2.$$

Summing (5.5) over these indices gives

$$r^{-2}\int_{B(x,r)} T_n \wedge \beta \le C_R\left(\frac{\operatorname{osc}_{B(x,r)} \rho}{b-a} + \frac{2}{n}\right). \tag{5.6}$$

Multiply β by a cutoff equal to one on $B(x,R)$. The corresponding positive measures μ_n converge weakly to μ. For the open ball, the Portmanteau inequality and (5.6) give

$$\mu(B(x,r)) \le \liminf_n \mu_n(B(x,r)) \le \frac{C_R r^2}{b-a}(M_r - m_r).$$

Consequently,

$$r^{-2}\int_{B(x,r)} T \wedge \beta \le \frac{C_R}{b-a}\operatorname{osc}_{B(x,r)} \rho \longrightarrow 0 \qquad (r \downarrow 0). \tag{5.7}$$

Thus $\nu(T,x) = 0$ on X. Since $\operatorname{supp} T \Subset X$, the Lelong numbers vanish on all of S. □

Lemma 5.6. *For a positive dd^c-closed $(1,1)$-current T on a compact Kähler surface with all Lelong numbers zero, there is a class $\alpha_T \in H^{1,1}(S,\mathbb{R})$ satisfying*

$$\alpha_T \cdot [\eta] = \int_S T \wedge \eta \qquad (d\eta = 0), \tag{5.8}$$

for every smooth real $(1,1)$-form η. Moreover,

$$\alpha_T^2 \ge 0, \qquad dT \ne 0 \implies \alpha_T^2 > 0. \tag{5.9}$$

If $dT = 0$, then $\alpha_T = [T]$.

Proof. The decomposition [DNS22, §2.1 and Lemma B.2]

$$T = \Omega + \partial R + \bar\partial\overline{R} + i\partial\bar\partial u$$

has Ω smooth, real, and closed, R of bidegree $(0,1)$, and u a real distribution. For $d\eta = 0$, its pure bidegree implies $\partial\eta = \bar\partial\eta = 0$. Thus

$$\langle \partial R, \eta\rangle = \langle \bar\partial\overline{R}, \eta\rangle = \langle i\partial\bar\partial u, \eta\rangle = 0,$$

which proves (5.8) with $\alpha_T = [\Omega]$. The nondegenerate intersection pairing on $H^{1,1}(S,\mathbb{R})$ makes the class independent of the decomposition. The density formula [DNS22, Theorem 2.2(2), Corollary 2.4] gives

$$\alpha_T^2 = \|\vartheta\| + 2E(T), \qquad \|\vartheta\| \geq 0, \quad E(T) \geq 0, \quad E(T) = 0 \Longleftrightarrow dT = 0.$$

Here ϑ is the positive measure on the diagonal determined by a tangent current of $T \otimes T$, and $E(T) = \int_S \bar{\partial} R \wedge \partial \overline{R}$ is the current energy, independent of the chosen decomposition. The hypotheses of the density formula hold because all Lelong numbers vanish. This proves (5.9). □

Proof of Theorem 5.4. If $r : X \to Y$ is a proper modification of a Stein space, the pullback of a smooth strictly psh exhaustion of Y is the required exhaustion. Otherwise the preceding lemmas give a current T with support in $K \Subset X$ and a class $\alpha = \alpha_T$ satisfying

$$\alpha \cdot [\omega] = 1, \qquad \alpha^2 \geq 0, \qquad dT \neq 0 \Longrightarrow \alpha^2 > 0. \tag{5.10}$$

For each component D_i, choose a metric of curvature θ_i and canonical section s_i. Choose $\chi \in C_c^\infty(X)$ equal to one near K, and extend $v_i = \chi \log \|s_i\|^2$ by zero to S. Since s_i is nonvanishing near K, Poincaré–Lelong gives $dd^c v_i = -\theta_i$ there. Therefore

$$\alpha \cdot [D_i] = \int_S T \wedge \theta_i = -\langle T, dd^c v_i \rangle = 0. \tag{5.11}$$

For a connected boundary component E, let λ_E be the largest eigenvalue of its intersection matrix I_E. Lemma 5.1(i) excludes $\lambda_E < 0$. If $\lambda_E > 0$, rational approximation gives an integral divisor $B = \sum b_i D_i$, with all $b_i > 0$ and $B \cdot D_i > 0$ on E. In particular, $B^2 = \sum b_i (B \cdot D_i) > 0$. Then $\alpha \cdot [B] = 0$, and Hodge index forces $\alpha^2 < 0$, contrary to (5.10). Therefore $\lambda_E = 0$. Distinct irreducible curves have nonnegative intersection, and connectedness makes I_E irreducible. Perron–Frobenius gives a one-dimensional kernel with a positive rational generator. Choosing a positive integral generator for each E and summing gives

$$A = \sum_i a_i D_i, \qquad a_i > 0, \qquad \operatorname{supp} A = D, \qquad A \cdot D_i = A^2 = 0. \tag{5.12}$$

Set $c = (\int_A \omega)^{-1}$ and $v = \alpha - c[A]$. Effectiveness gives $\int_A \omega > 0$, while (5.10)–(5.11) give

$$v \cdot [\omega] = 0, \qquad v \cdot [A] = 0, \qquad v^2 = \alpha^2 \geq 0.$$

Hodge index is negative definite on $[\omega]^\perp$, so $v = 0$ and $\alpha^2 = 0$. The strict inequality in (5.10) then forces $dT = 0$. We have proved

$$dT = 0, \qquad [T] = \alpha = c[A], \qquad c = \left(\int_A \omega \right)^{-1} > 0. \tag{5.13}$$

The dd^c-lemma for currents now supplies a real distribution u with

$$dd^c u = T - c[A]. \tag{5.14}$$

Choose a neighborhood U of D disjoint from K. On a trivializing chart $U_i \subset U$, write $s_A = f_i e_i$. Poincaré–Lelong and (5.14) give the distributional identity

$$dd^c (u + c \log |f_i|^2) = -c[A] + c[A] = 0.$$

Elliptic regularity for the pluriharmonic distribution $u + c \log |f_i|^2$ gives

$$u = -c \log |f_i|^2 + h_i, \qquad h_i \text{ smooth and pluriharmonic on } U_i. \tag{5.15}$$

The functions $-c \log |f_i|^2 + h_i$ agree on overlaps off D, since they represent the same distribution u. They therefore define a smooth pluriharmonic function on $U \setminus D$. In a finite relatively

compact chart cover of D, each h_i is bounded and f_i tends to zero along D. Since every $a_i > 0$, the function tends uniformly to $+\infty$ along D. Lemma 5.1(ii) gives the exhaustion.

For the metric statement, set $b_i = u/c + \log|f_i|^2 = h_i/c$ on U_i. If $e_j = g_{ij}e_i$, then $f_j = g_{ij}^{-1}f_i$ and

$$b_j = b_i - \log|g_{ij}|^2, \qquad dd^c b_i = 0.$$

Thus $\|e_i\|^2_{h_{\mathrm{flat}}} = e^{-b_i}$ defines a smooth flat Hermitian metric on $\mathcal{O}_S(A)|_U$, with $-\log\|s_A\|^2_{h_{\mathrm{flat}}} = u/c$. For a global smooth metric h_0 with weights b_i^0, the differences $b_i - b_i^0$ define a single smooth function b on U. Choose a cutoff χ supported in U and equal to one on a smaller neighborhood of D. The global metric $h = e^{-\chi b}h_0$ agrees with h_{flat} there, proving the claimed curvature statement. □

Corollary 5.7. *Every smooth complex quasi-projective surface with a continuous psh exhaustion has a smooth psh exhaustion.*

Proof. Resolve a projective closure and principalize the complement ideal, with centers outside X [Wło05, Theorems 1.0.1–1.0.2]. The resulting morphism $p : S \to \overline{X}$ is an isomorphism over X, and $D = (p^{-1}(\overline{X} \setminus X))_{\mathrm{red}}$ is SNC. Thus p identifies $S \setminus D$ with X. Apply Theorem 5.4 to each noncompact component. Compact components need only constant functions, and there are finitely many components. □

For an example with $\mathcal{O}(X) = \mathbb{C}$, take a compact Riemann surface B of positive genus and a non-torsion $A \in \mathrm{Pic}^0(B)$. The squared unitary-flat norm is a smooth psh exhaustion of $X = \mathrm{Tot}(A^{-1})$. The degree-n fiberwise Taylor coefficient of a holomorphic function on X belongs to $H^0(B, A^n) = 0$ for $n > 0$, and its constant coefficient is constant on B.

Corollary 5.8. *Let $C \subset S$ be a smooth connected square-zero curve in a compact Kähler surface. A continuous divergent psh collar at C implies a smooth psh exhaustion of $S \setminus C$. If the complement is not a modification of a Stein space, $\mathcal{O}_S(C)$ is unitary flat near C, and C has a pluriharmonic divergent collar.*

Proof. Let Φ be the collar, choose a neighborhood W of C such that Φ is defined near $\overline{W} \setminus C$, and take $M > \sup_{\partial W}\Phi$. Set

$$\rho(x) = \begin{cases} \max\{\Phi(x), M\}, & x \in W \setminus C, \\ M, & x \notin W. \end{cases}$$

The formulas agree near ∂W, and divergence at C makes ρ a continuous psh exhaustion. In Theorem 5.4, the single-component kernel generator can be chosen to be 1, so $A = C$. □

5.3. **Numerical existence criteria.** We next use the classification of compact surfaces to obtain a numerical criterion for nonprojective surfaces, followed by the corresponding criterion for smooth ruled surfaces.

Theorem 5.9. *Let S be a smooth compact connected nonprojective surface and $D \neq 0$ an effective divisor. The following conditions are equivalent.*

(i) *$S \setminus \operatorname{supp} D$ admits a continuous psh exhaustion.*
(ii) *$S \setminus \operatorname{supp} D$ admits a smooth psh exhaustion.*
(iii) *no connected boundary component has negative definite intersection matrix.*

Proof. The implication (ii)⇒(i) is immediate, and Lemma 5.1(i) gives (i)⇒(iii). To prove (iii)⇒(ii), we distinguish the two possible values of the algebraic dimension $a(S)$, using the classification in [FN05, §2].

If $a(S) = 1$, the algebraic reduction is a holomorphic elliptic fibration $f : S \to B$ with connected fibers, and every curve is vertical. For a fiber $F = \sum m_i C_i$, the identities $F \cdot C_i = 0$ imply $m_i C_i^2 = -\sum_{j \neq i} m_j (C_i \cdot C_j)$. Substitution gives

$$x^{\mathsf{T}} I x = -\sum_{i<j} (C_i \cdot C_j) m_i m_j \left(\frac{x_i}{m_i} - \frac{x_j}{m_j} \right)^2, \qquad I = (C_i \cdot C_j). \tag{5.16}$$

The right-hand side vanishes precisely when all x_i/m_i are equal, since the fiber is connected. Extend a vector on a proper subset of the components by zero. Equality then forces the entire vector to vanish. Thus every proper principal submatrix is negative definite. Condition (iii) implies that the boundary contains each fiber support it meets, and $X = f^{-1}(B \setminus P)$ for a finite nonempty P. For a smooth subharmonic exhaustion τ of $B \setminus P$, $\tau \circ f$ is psh and $\{\tau \circ f \leq c\} = f^{-1}(\{\tau \leq c\})$ is compact.

Suppose $a(S) = 0$. By [FN05, Lemma 2.1], $c_1(L)^2 \leq 0$ for every line bundle L. Perron–Frobenius and (iii) therefore give an effective A of full boundary support with $A^2 = 0$. Indeed, a positive largest eigenvalue would give an effective divisor of positive square, and a negative one is excluded by (iii). At a blowdown $q : T \to T'$ with exceptional curve E, put $A' = q_* A$ and write $A = q^* A' + bE$. Since $q^* A' \cdot E = 0$ and $E^2 = -1$,

$$0 = A^2 = (A')^2 - b^2, \qquad (A')^2 \leq 0,$$

so $b = 0$ and $A = q^* A'$. Iterating to the minimal model $p : S \to S_0$ gives $A = p^* A_0 \neq 0$, $A_0^2 = 0$, and $\operatorname{supp} A = p^{-1}(\operatorname{supp} A_0)$. The restriction

$$p : S \setminus \operatorname{supp} A \longrightarrow S_0 \setminus \operatorname{supp} A_0$$

is proper because the deleted set is the full inverse image of $\operatorname{supp} A_0$. Pullback therefore preserves an exhaustion.

If S_0 is Kähler, it is a torus or K3 of algebraic dimension zero. On a torus, translating an irreducible curve shows its square is nonnegative, hence zero. Adjunction then gives arithmetic genus one. A rational normalization is impossible, since any map from $\mathbf{P}^1$ lifts to $\mathbb{C}^2$ and is constant. The curve is therefore a smooth elliptic curve, and a translate is an elliptic subtorus. The quotient onto the one-dimensional torus gives a nonconstant meromorphic function. On a K3, $h^0(\mathcal{O}(-A_0)) = 0$ because A_0 is effective and nonzero, while Riemann–Roch and Serre duality give

$$h^0(\mathcal{O}(A_0)) - h^1(\mathcal{O}(A_0)) = 2 + \tfrac{1}{2} A_0^2 = 2.$$

Two independent sections have a nonconstant meromorphic ratio. Both possibilities contradict $a(S_0) = 0$.

The remaining minimal surfaces are of class VII. If $b_2(S_0) = n > 0$, Enoki's square-zero-divisor theorem [Eno81, Main Theorem, p. 454] makes S_0 an Enoki surface and gives

$$A_0 = m C_{\mathrm{E}}, \qquad m \in \mathbb{Z}_{>0}, \qquad \operatorname{supp} A_0 = C_{\mathrm{E}},$$

where C_{E} is its distinguished reduced cycle. By [Eno81, Proposition 3.5(iii), p. 460], $\mathcal{A} = S_0 \setminus C_{\mathrm{E}}$ is an affine-line bundle over an elliptic curve B_{E}. Its linearization Q has $\deg Q = -n$, using the affine-bundle degree convention of [Eno81, §2, p. 457].

Write its affine coordinate changes as $w_i = a_{ij} w_j + b_{ij}$. Extending these maps to $\mathbf{P}^1$ gives a compact ruled surface $\overline{\mathcal{A}}$ with infinity section Y_∞. In the coordinates $u_i = w_i^{-1}$ near this section,

$$u_i = \frac{u_j}{a_{ij} + b_{ij} u_j} = a_{ij}^{-1} u_j + O(u_j^2), \qquad N_{Y_\infty / \overline{\mathcal{A}}} \simeq Q^{-1}.$$

Consequently

$$Y_\infty^2 = \deg Q^{-1} = n > 0, \qquad \overline{\mathcal{A}} \setminus Y_\infty \simeq S_0 \setminus \operatorname{supp} A_0.$$

Lemma 5.1(iii) supplies a smooth strictly psh divergent collar at Y_∞, and (ii) gives a smooth psh exhaustion of its complement. If $b_2(S_0) = 0$, a minimal class VII surface containing a curve is elliptic or Hopf [Eno81, p. 453]. Since $a(S_0) = 0$, it is Hopf. Since $b_2 = 0$, every curve has square zero and distinct curves have intersection zero. Adjunction gives arithmetic genus one, while rational normalizations are impossible because maps $\mathbf{P}^1 \to S_0$ lift to $\mathbb{C}^2 \setminus \{0\}$ and are constant. Thus the curves are disjoint smooth elliptic curves. Finite Ueda type gives a smooth strictly psh collar [CLPT18, Theorem 2.5]. Infinite type gives a unitary flat neighborhood by [CFS+26, Theorem 1.6], hence a pluriharmonic collar. Glue the collars. □

Theorem 5.10. *Let $f : S \to B$ be a smooth $\mathbf{P}^1$-bundle over a compact Riemann surface and let $D \neq 0$ be a reduced SNC divisor on S. The three conditions in Theorem 5.9 are equivalent.*

Proof. As above, it suffices to construct smooth divergent collars under condition (iii). Lemma 5.1 handles boundary components with a positive eigenvalue. For a connected component E with zero largest eigenvalue, choose its positive kernel divisor A. If E is vertical, it is a whole fiber, and the punctured-base construction applies. Otherwise, for a fiber F of f, put $a = A \cdot F > 0$. A vertical component would be a fiber and have intersection a with A, contradicting $A \cdot E_i = 0$. Thus every E_i is horizontal. Choose a section C_0 of the projective ruling and write $C_0^2 = -e$. Write

$$A \equiv aC_0 + bF, \qquad E_i \equiv d_iC_0 + b_iF, \qquad d_i = E_i \cdot F > 0.$$

Using $C_0 \cdot F = 1$ and $F^2 = 0$, compute

$$0 = A^2 = -ea^2 + 2ab, \qquad 0 = A \cdot E_i = ab_i + d_ib - ead_i.$$

Since $a > 0$, these identities give

$$b = ae/2, \qquad E_i \equiv d_i(C_0 + (e/2)F), \qquad E_i^2 = E_i \cdot E_j = 0 \quad (i \neq j).$$

Distinct components are therefore disjoint, and connectedness makes $E = C$ a single curve, smooth by the SNC hypothesis. For $d = C \cdot F$, the canonical class formula

$$K_S \equiv -2C_0 + (2g(B) - 2 - e)F$$

and adjunction give $2g(C)-2 = (K_S+C)\cdot C = d(2g(B)-2)$. The nonconstant map $f|_C : C \to B$ is finite of degree d. Riemann–Hurwitz now gives degree zero for its effective ramification divisor, so it is étale.

For $d = 1$, the complement $S \setminus C$ is an affine-line bundle with degree-zero linearization. In unitary flat frames write its transitions as $w_i = a_{ij}w_j + b_{ij}(z)$ with $|a_{ij}| = 1$. The construction of [KU25, Proposition 2.1] gives pluriharmonic functions h_i with $h_i = a_{ij}h_j + b_{ij}$. On simply connected charts split $h_i = f_i + g_i$, with f_i holomorphic and g_i antiholomorphic. The coordinate changes $w_i \mapsto w_i - f_i$ replace b_{ij} by

$$c_{ij} = b_{ij} + a_{ij}f_j - f_i = g_i - a_{ij}g_j.$$

This expression is both holomorphic and antiholomorphic, hence constant. In the new coordinates,

$$w_i = a_{ij}w_j + c_{ij}, \qquad g_i = a_{ij}g_j + c_{ij}, \qquad |a_{ij}| = 1.$$

The identity $w_i - g_i = a_{ij}(w_j - g_j)$ makes $\rho = |w_i - g_i|^2$ global. Since g_i is antiholomorphic,

$$\rho_{z\bar{z}} = |(g_i)_{\bar{z}}|^2, \qquad \rho_{z\bar{w}_i} = \rho_{w_i\bar{z}} = 0, \qquad \rho_{w_i\bar{w}_i} = 1.$$

Thus ρ is smooth and psh. On a finite relatively compact trivializing cover of the compact base, the g_i are bounded, so $\rho \leq c$ bounds each fiber coordinate. Its sublevels are compact.

For $d \geq 2$, let $B' \to B$ be the finite étale Galois cover associated to the kernel of the permutation monodromy $\pi_1(B) \to \mathfrak{S}_d$ of $C \to B$. Form

$$S' = S \times_B B', \qquad p : S' \to S, \qquad C' = p^{-1}(C) = C_1 \sqcup \cdots \sqcup C_d.$$

The C_i are sections with square zero because their normal bundles are pullbacks of $N_{C/S}$ along finite covers. If $d = 2$, these sections identify the ruling with $\mathbf{P}(\mathcal{O}_{B'} \oplus L)$, where $\deg L = 0$, and $S' \setminus C' = L^\times$. For a unitary flat metric, put $t(v) = \log \|v\|^2$. Then $dd^c t = 0$ on $L^\times$, and

$$\mathcal{L}_{t^2}(\xi) = 2|\partial t(\xi)|^2 \geq 0, \qquad \{t^2 \leq c\} = \{e^{-\sqrt{c}} \leq \|v\|^2 \leq e^{\sqrt{c}}\}$$

is compact. Hence t^2 is a smooth psh exhaustion.

For $d \geq 3$, the unique fiberwise Möbius map sending C_1, C_2, C_3 to $0, 1, \infty$ identifies S' with $B' \times \mathbf{P}^1$. Every remaining section is a holomorphic map $B' \to \mathbf{P}^1 \setminus \{0, 1, \infty\}$, which is constant by the maximum principle. Thus $S' \setminus C' = B' \times (\mathbf{P}^1 \setminus Q)$ for a finite set Q, and an exhaustion of the punctured fiber pulls back to one of the product.

For $d \geq 2$, let $\rho' \geq 0$ be the exhaustion just constructed and G the deck group of $B' \to B$. The function

$$\widehat{\rho} = \frac{1}{|G|} \sum_{g \in G} \rho' \circ g$$

is smooth, psh, and G-invariant. Since $\widehat{\rho} \geq \rho'/|G|$, its sublevels are compact. It descends under the finite étale map p to a smooth psh exhaustion of $S \setminus C$. Restrict this function to a collar at E and apply Lemma 5.1(ii) to glue all boundary collars. □

5.4. **Square-zero curves.** For a smooth square-zero curve, Ueda theory relates the formal neighborhood to the existence of holomorphic defining functions $w_i = t_{ij} w_j$ with $|t_{ij}| = 1$. These functions give a pluriharmonic collar through $-\log |w_i|^2$.

Proposition 5.11. *Let $C \subset S$ be smooth, connected, and square-zero, with S compact Kähler. Put $N = N_{C/S}$, let $\chi : H_1(C, \mathbb{Z}) \to \mathrm{U}(1)$ be its unitary character, and set $K = \ker[H_1(C, \mathbb{Z}) \to H_1(S, \mathbb{Z})]$. The complement $S \setminus C$ has a smooth psh exhaustion if any of the following conditions holds.*

(i) *C has finite Ueda type or a unitary flat neighborhood.*
(ii) *N is torsion or Diophantine.*
(iii) *$\chi(K)$ is finite.*
(iv) *$q(S) \geq g(C)$.*
(v) *$g(C) = 1$.*

Here Diophantine means $\operatorname{dist}(N^m, \mathcal{O}_C) \geq cm^{-\tau}$ for some $c, \tau > 0$ and all $m \geq 1$, with respect to a fixed flat metric on $\mathrm{Pic}^0(C)$. We write $q(S) = h^1(S, \mathcal{O}_S)$ for the irregularity.

Proof. Finite type gives a smooth strictly psh collar [CLPT18, Theorem 2.5]. In a unitary flat neighborhood, $w_i = t_{ij} w_j$ with $|t_{ij}| = 1$, so the functions $-\log |w_i|^2$ agree and form a pluriharmonic divergent collar. In infinite type, Ueda's convergence theorem [Ued83, Theorem 3] supplies such a neighborhood when N is torsion or Diophantine.

Suppose $\chi(K)$ is finite, and let k annihilate this finite group. Write $j_* : H_1(C, \mathbb{Z}) \to H_1(S, \mathbb{Z})$ for the inclusion map. The formula

$$\chi_0(j_* \gamma) = \chi(\gamma)^k$$

defines a character on $\operatorname{im} j_*$, because two lifts differ by an element of K. Divisibility of $\mathrm{U}(1)$ extends it to a character $\widetilde{\chi} : H_1(S, \mathbb{Z}) \to \mathrm{U}(1)$, with $\widetilde{\chi} \circ j_* = \chi^k$. For non-torsion N of infinite type, [CLPT18, Theorem 4.4] then gives an effective divisor E disjoint from C with $[E] = m[C]$

for some $m > 0$. The line bundle $\mathcal{O}_S(mC - E)$ has zero real Chern class and hence a unitary flat metric on the compact Kähler surface. Its canonical meromorphic section $\sigma = s_C^m / s_E$ satisfies

$$\operatorname{div}(\sigma) = mC - E, \qquad dd^c \log \|\sigma\|^2 = m[C] - [E].$$

On a neighborhood of C disjoint from E, the function $-m^{-1} \log \|\sigma\|^2$ is therefore pluriharmonic off C and equals $-\log |w|^2$ plus a smooth function. It is the required divergent collar.

For condition (iv), consider the restriction $r : H^1(S, \mathcal{O}_S) \to H^1(C, \mathcal{O}_C)$. If r is not injective, [CLPT18, Proposition 2.7] makes a multiple of C a fiber, and the punctured-base construction applies. Otherwise, since $q(S) \geq g(C)$, injectivity makes r an isomorphism. Hodge decomposition and complex conjugation then give an isomorphism $H^1(S, \mathbb{Q}) \to H^1(C, \mathbb{Q})$, whose dual is $j_* \otimes \mathbb{Q}$. Consequently $K \otimes \mathbb{Q} = 0$. As a subgroup of the free abelian group $H_1(C, \mathbb{Z})$, K is torsion-free, hence $K = 0$, and the preceding character argument applies. Finally, for (v), [CFS+26, Theorem 1.6] shows that an elliptic curve of infinite Ueda type has a unitary flat neighborhood. Apply Lemma 5.1(ii) in all cases. □

6. Holomorphic convexity for curve fiberings

Complex spaces in this section are reduced, Hausdorff, and countable at infinity. A smooth function is locally the restriction of a smooth function in an ambient complex manifold. Fiber components refer to their reduced supports.

Theorem 6.1 (=Theorem 1.4)**.** *Let X have pure dimension $n \geq 2$ and admit a continuous real-valued psh exhaustion ρ. If a holomorphic map*

$$F : X \longrightarrow \mathbb{C}^{n-1} \quad \textit{satisfies} \quad \dim F^{-1}(z) \leq 1 \qquad (z \in F(X)),$$

then X is holomorphically convex and admits a smooth psh exhaustion.

For smooth X, the convexity argument follows [Vîj22b, Corollary 4.2 and pp. 463–464]. For singular X, we first control compact fiber components using analytic cycles. This allows us to choose exhaustion shells whose fibers are Stein and then apply the same convexity criteria.

6.1. **Compact fiber components.** Write $F = (f_1, \dots, f_{n-1})$. On each local irreducible branch of X at x, the fiber through x is defined by the $n-1$ equations $f_k = f_k(x)$. The analytic dimension theorem therefore gives

$$\dim_x F^{-1}(F(x)) \geq n - (n-1) = 1.$$

Together with the upper bound in the theorem, this proves that F is equidimensional with one-dimensional fibers on every local branch. Its graph and projection are

$$\iota : X \longrightarrow \Gamma_F \subset \mathbb{C}^{n-1} \times X, \qquad x \longmapsto (F(x), x), \qquad p : \Gamma_F \longrightarrow \mathbb{C}^{n-1}.$$

Since ι is an isomorphism and $p \circ \iota = F$, the reduced graph, with multiplicity one on each component, defines an analytic family of geometric fiber cycles over the normal base $\mathbb{C}^{n-1}$ [BM98, Chapter I, §1(b), p. 816]. Write this family as

$$\mathcal{F}_z = \sum_j m_j(z)[C_j(z)], \qquad m_j(z) \in \mathbb{Z}_{>0}, \qquad |\mathcal{F}_z| = F^{-1}(z). \tag{6.1}$$

The fiber cycles are locally finite, with the zero cycle over an empty fiber.

Lemma 6.2 (Uniform area)**.** *Fix a Hermitian metric h on X. Given compact $L \subset X$ and $z_0 \in \mathbb{C}^{n-1}$, there are a neighborhood B of z_0 and $A > 0$ such that*

$$\int_C h \leq A$$

for every compact irreducible component $C \subset L$ of a fiber over B.

Proof. Choose $\chi \in C_c^\infty(X)$ with $0 \le \chi \le 1$ and $\chi = 1$ near L. The map

$$I : \mathbb{C}^{n-1} \longrightarrow \mathbb{R}_{\ge 0}, \qquad I(z) = \int_{\mathcal{F}_z} \chi h$$

is continuous by the theorem on integration of compactly supported forms over analytic families of cycles [BM14, Chapter IV, Proposition 2.3.1]. Choose B so that $I(z) < I(z_0) + 1$ on B and set $A = I(z_0) + 1$. If $C = C_j(z) \subset L$, its multiplicity in (6.1) is positive. Hence

$$\int_C h = \int_C \chi h \le \int_{\mathcal{F}_z} \chi h = I(z) < A.$$

□

Lemma 6.3 (Stability)**.** *Let $K \subset X$ be compact. If every irreducible component of $F^{-1}(z_0)$ meeting K is noncompact, the same holds for all fibers over a neighborhood of z_0.*

Proof. Suppose the conclusion fails. There are $z_j \to z_0$, compact irreducible components $C_j \subset F^{-1}(z_j)$, and $x_j \in C_j \cap K$. After taking a subsequence, $x_j \to x \in K$. The maximum principle on the normalization of C_j makes $\rho|_{C_j}$ constant. Thus, with $R = \max_K \rho$,

$$\rho|_{C_j} = \rho(x_j) \le R, \qquad C_j \subset L := \{\rho \le R\} \Subset X.$$

Lemma 6.2 bounds $\int_{C_j} h$ for all sufficiently large j. Bishop compactness for cycles on complex spaces [Bar99, Theorem 1] yields a convergent subsequence

$$[C_j] \longrightarrow T \quad \text{in the space of compact effective one-cycles of } X.$$

Convergence of compact cycles gives upper semicontinuity of their supports, so $x_j \to x$ implies $x \in |T|$. Convergence of the associated currents gives $|T| \subset L$. Moreover, a point outside $F^{-1}(z_0)$ has a neighborhood disjoint from $F^{-1}(z_j)$ for all large j, so it is outside $|T|$. Therefore

$$x \in |T| \subset L \cap F^{-1}(z_0).$$

An irreducible component of $|T|$ through x is a compact curve in the pure one-dimensional fiber $F^{-1}(z_0)$ and hence a component of that fiber. It meets K, contradicting the hypothesis. □

6.2. **Convexity and reduction.** We use the following three criteria for complex spaces [Vîj22b, Propositions 2.2, 2.4 and Corollary 2.5].

(i) A complex space with a continuous psh function u is holomorphically convex if and only if every $\{u < t\}$ is.
(ii) A space with a continuous psh exhaustion is holomorphically convex if it is locally holomorphically convex over a Stein base.
(iii) If $\pi : M \to S$ is holomorphic, S is Stein, and an open set $D \subset M$ admits a continuous psh exhaustion, then D is holomorphically convex provided $\overline{D}^M \to S$ is proper and a neighborhood of $\partial_M D$ has Stein reduced fibers.

Here local holomorphic convexity over S means that every point of S has a neighborhood V whose inverse image is holomorphically convex.

For (iii), the Stein-neighborhood construction in the cited proof gives, over a relatively compact base neighborhood $V \Subset S$, finitely many smooth functions h_j and nonnegative base cutoffs τ_j for which

$$h_0 = \sum_j (\tau_j \circ \pi) h_j, \qquad h = h_0 + \Lambda\phi \circ \pi$$

is strictly psh on a collar W of the boundary over $\overline{V}$, where ϕ is strictly psh on the base. The product rule and Cauchy–Schwarz on the finitely many local embeddings give constants $a, b, c > 0$ such that

$$\mathcal{L}_{h_0}(\xi) \geq a\|\xi\|^2 - b\|d\pi(\xi)\|^2, \qquad \mathcal{L}_{\phi\circ\pi}(\xi) \geq c\|d\pi(\xi)\|^2.$$

Taking $\Lambda c > b$ makes h strictly psh on the collar. The functions are smooth on a neighborhood of $\overline{D} \cap \pi^{-1}(\overline{V})$. Let η_D be a continuous psh exhaustion of D and put

$$\psi = \eta_D + h \quad \text{on } D \cap \pi^{-1}(V).$$

The set

$$H = \left(\overline{D} \cap \pi^{-1}(\overline{V})\right) \setminus W$$

is a compact subset of D. Choose a constant A larger than $\max_H \psi$ when H is nonempty. Then $\{\psi > A\} \subset W$, where ψ is strongly psh. For every compact $T \subset V$, the function h has a lower bound m_T on $\overline{D} \cap \pi^{-1}(T)$, and

$$\{\psi \leq t\} \cap \pi^{-1}(T) \subset \{\eta_D \leq t - m_T\} \cap \pi^{-1}(T) \Subset D.$$

The set on the left is closed in the compact set on the right. Thus ψ is a vertical exhaustion and defines a 1-convex map. Holomorphic convexity follows from [Vîj22b, Theorem 2.3 and Proposition 2.4].

Lemma 6.4. *For every $\lambda \in \mathbb{R}$ and $z_0 \in \mathbb{C}^{n-1}$, there are $c > \lambda$ and a ball $B = B_r(z_0)$ such that*

$$D = F^{-1}(B) \cap \{\rho < c\} \quad \textit{and} \quad F^{-1}(B) \cap \{\rho < \lambda\}$$

are holomorphically convex.

Proof. Let $\mathcal{V}_0$ be the set of constant ρ-values on compact irreducible components of $F^{-1}(z_0)$. For each R, every component whose value is at most R lies in the compact set $\{\rho \leq R\}$. The locally finite family of irreducible components of the central fiber has only finitely many members meeting this set. Therefore

$$\#(\mathcal{V}_0 \cap (-\infty, R]) < \infty.$$

Choose $\lambda < a < c < b$ with $[a, b] \cap \mathcal{V}_0 = \varnothing$ and put $K = \{a \leq \rho \leq b\}$. This is compact, and no compact component of $F^{-1}(z_0)$ meets it. Lemma 6.3 gives a ball $B = B_r(z_0)$ such that no compact component of a fiber over B meets K. Set

$$M = F^{-1}(B), \qquad D = M \cap \{\rho < c\}, \qquad U = M \cap \{a < \rho < b\}.$$

Continuity of ρ gives $\partial_M D \subset \{\rho = c\} \cap M \subset U$. If a reduced fiber of $F|_U$ had a compact irreducible component C, compactness would make C a closed analytic subset of X. Since it has dimension one, it would be a component of the full F-fiber. This contradicts the choice of B, because $C \subset K$. Hence every reduced U-fiber is Stein by the criterion for curves without compact irreducible components [Vîj22a, §3, p. 1663].

On D, define

$$\eta = \max\left\{\frac{1}{c - \rho}, \frac{1}{r^2 - \|F - z_0\|^2}\right\}.$$

The function $t \mapsto (\alpha - t)^{-1}$ is convex and increasing on $t < \alpha$, so both terms and their maximum are continuous and psh. For $t > 0$,

$$\{\eta \leq t\} = \{\rho \leq c - 1/t\} \cap F^{-1}\{\|z - z_0\|^2 \leq r^2 - 1/t\} \Subset D.$$

Thus η is a continuous psh exhaustion of D. For compact $T \subset B$,

$$\overline{D}^{\,M} \cap F^{-1}(T) \subset \{\rho \leq c\} \cap F^{-1}(T).$$

The left-hand side is closed in the compact set on the right, so $\overline{D}^M \to B$ is proper. Criterion (iii) now gives holomorphic convexity of D. Since $\lambda < c$,

$$F^{-1}(B) \cap \{\rho < \lambda\} = \{x \in D : \rho(x) < \lambda\},$$

and criterion (i) gives the second assertion. □

Proof of Theorem 6.1. For $X(\lambda) = \{\rho < \lambda\}$, Lemma 6.4 makes $F|_{X(\lambda)} : X(\lambda) \to \mathbb{C}^{n-1}$ locally holomorphically convex. The continuous psh function $\eta_\lambda = (\lambda - \rho)^{-1}$ exhausts $X(\lambda)$, since

$$\{\eta_\lambda \le t\} = \{\rho \le \lambda - 1/t\} \Subset X(\lambda) \qquad (t > 0).$$

Criterion (ii) makes every $X(\lambda)$ holomorphically convex, and criterion (i) then applies to ρ on X. Let $q : X \to Z$ be the proper Remmert reduction onto its Stein space Z, and let u be a smooth strictly psh exhaustion of Z. Holomorphic pullback preserves smoothness in local embeddings and plurisubharmonicity, while properness gives

$$u \circ q \in C^\infty(X) \cap \mathrm{PSH}(X), \qquad \{u \circ q \le t\} = q^{-1}(\{u \le t\}) \Subset X.$$ □

Corollary 6.5 (Simultaneous contraction)**.** *The Remmert reduction $q : X \to Z$ has connected fibers, $q_*\mathcal{O}_X = \mathcal{O}_Z$, and $F = \widetilde{F} \circ q$ for a holomorphic $\widetilde{F} : Z \to \mathbb{C}^{n-1}$. Its nontrivial fibers are exactly the maximal connected unions of compact irreducible fiber curves of F.*

Proof. By $q_*\mathcal{O}_X = \mathcal{O}_Z$, each coordinate f_k of F descends to a holomorphic function $\widetilde{f_k}$ on Z. These functions define $\widetilde{F}$ with $F = \widetilde{F} \circ q$. Every q-fiber is therefore a compact connected analytic subset of an F-fiber. If it is not a point, the dimension bound and local finiteness make it a finite connected union of compact irreducible components of that F-fiber.

Conversely, the image of a compact irreducible curve in the Stein space Z is a point, so q contracts each such curve. Consider a connected union E of compact irreducible components of one F-fiber. The normalization maximum principle makes ρ constant on each component. Intersecting components have the same value. By local finiteness and connectedness, any two components can be joined by a finite chain of intersecting components. Thus $\rho|_E = a$ for one $a \in \mathbb{R}$, so $E \subset \{\rho \le a\}$. Only finitely many fiber components meet this compact set. Hence E is a finite connected union and q is constant on E. It follows that the maximal such unions are precisely the nontrivial q-fibers. □

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